Reference papers

Hepatic and liver vessel segmentation (primary focus)

1. **3D Graph-Connectivity Constrained Network for Hepatic Vessel Segmentation**

- Authors: Ruikun Li, Yi-Jie Huang, Huai Chen, Xiaoqing Liu, Yizhou Yu, Dahong Qian, Lisheng Wang
- Year: 2022 · Source: IEEE Xplore (IEEE J. Biomed. Health Informatics 26(3))
- URL: https://ieeexplore.ieee.org/document/9562259
- Relevance: Integrates a graph-attention module into a 3D U-Net to enforce vascular connectivity arXiv +2 on MSD Task08 — a direct, high-quality baseline and topological motivation parallel to clDice.

Abstract

Segmentation of hepatic vessels from 3D CT images is necessary for accurate diagnosis and preoperative planning for liver cancer. However, due to the low contrast and high noises of CT images, automatic hepatic vessel segmentation is a challenging task. Hepatic vessels are connected branches containing thick and thin blood vessels, showing an important structural characteristic or a prior: the connectivity of blood vessels. However, this is rarely applied in existing methods. In this paper, we segment hepatic vessels from 3D CT images by utilizing the connectivity prior. To this end, a graph neural network (GNN) used to describe the connectivity prior of hepatic vessels is integrated into a general convolutional neural network (CNN). Specifically, a graph attention network (GAT) is first used to model the graphical connectivity information of hepatic vessels, which can be trained with the vascular connectivity graph constructed directly from the ground truths. Second, the GAT is integrated with a lightweight 3D U-Net by an efficient mechanism called the plug-in mode, in which the GAT is incorporated into the U-Net as a multi-task branch and is only used to supervise the training procedure of the U-Net with the connectivity prior. The GAT will not be used in the inference stage, and thus will not increase the hardware and time costs of the inference stage compared with the U-Net. Therefore, hepatic vessel segmentation can be well improved in an efficient mode. Extensive experiments on two public datasets show that the proposed method is superior to related works in accuracy and connectivity of hepatic vessel segmentation.

Introduction

Liver cancer is one of the most common and deadliest malignant cancers across the globe, whose occurrence has been increasing for several years [1]. For liver cancer, according to different stages of the tumors and the complex vascular networks, different treatments such as surgical resection, ablation, and embolization can be used in clinical practice [2]. In these treatments, quantitative analysis and accurate modeling of hepatic vessels is needed, which can reveal the physiological form and structural characteristics of hepatic vessels, providing useful information for diagnosis and accurate preoperative planning [3], [4]. Thus, it is important to segment hepatic vessels from 3D computed tomography (CT) images. However, accurate segmentation of hepatic vessels is challenging because of the complex vascular structure, low contrast, and high noises in the CT images [5]. Manual delineation of hepatic vessels from CT images slice by slice is time-consuming and observer-dependent. Thereby automatic and accurate segmentation of hepatic vessels from CT images is under urgent demand.

Recently, deep learning-based methods have demonstrated powerful ability of feature extraction and analysis and achieved remarkable results in medical image analysis [6], [7]. Particularly, the convolutional neural networks (CNN) have been widely applied in medical image segmentation and have become the baseline for many tasks. For example, the 3D U-Net [8] and V-Net [9] are popular ones for segmenting various organs and lesions in 3D medical images. nnU-Net [10], which is a self-configuring method, has achieved remarkable performance in many segmentation tasks without manual intervention. Recently, different variants of U-Net have gradually become the basic paradigm of medical image segmentation [11]. However, these methods are not directly designed for segmenting hepatic vessels and do not make good use of the connectivity characteristics of hepatic vessels.

Some researchers have proposed deep learning-based methods for hepatic vessel segmentation [12]–[16]. Kitrungrotsakul *et al.* [17] proposed a novel architecture consisting of three parallel deep convolutional neural networks to generate voxel-wise prediction for hepatic vessel segmentation, in which the DenseNet [18] was used as the backbone for each pathway to extract features from three orthogonal planes of 3D CT images. Huang *et al.* [19] proposed a robust hepatic vessel segmentation method based on 3D U-Net and a variant dice loss, which improved segmentation accuracy and sensitivity with unbalanced foreground and background. At the same time, they also discussed the impact of the quality of hepatic vessel annotation on the evaluation results. Yan *et al.* [2] proposed an attention-guided concatenation module to better select and utilize deep features in a general U-Net architecture, and a multi-scale fusion block was further introduced to implicitly improve the linking of local vessel fragments. These methods mainly consider 3D context utilization and class imbalance, and their predictions usually can achieve high scores on voxel-wise metrics. However, their segmented vessels may be broken, discontinuous, wrongly crossing, and often lose many vessels with small diameters. Such vessels still cannot meet clinical requirements.

Human's organs usually have certain shape priors. Thus, researchers have recently integrated prior knowledge of organs into CNN to generate more reasonable and realistic segmentation results [20]–[24]. Hepatic vessels are some connected branches containing many thick and thin blood vessels. While they usually have individual variations in shape and size between different people, which makes their segmentation a difficult problem, they show common structural characteristics or priors: they are typical tree-like structures with both thick and thin blood vessels connected. Generally, maintaining the connectivity and topological structure of the segmented blood vessels is a clinical and human experience requirement, which should be considered in the design of the blood vessel segmentation algorithm. To our knowledge, however, the connectivity prior is rarely applied in existing deep learning methods for segmenting hepatic vessels. In this paper, we will incorporate the connectivity prior of hepatic vessels into CNN to improve hepatic vessel segmentation in 3D CT images.

Graph neural networks (GNN) can effectively analyze data generated from non-Euclidean domain [25], and it has been used to describe and analyze the connectivity and tree-like structure [26]–[29]. Juarez *et al.* [26] introduced a GNN module at the deepest level of the U-Net to improve the airway segmentation. However, the node connectivity defined in [26] does not exactly reflect the topological structure of the airway, and the GNN module cannot explicitly analyze and constrain the connectivity prior. In [27], a graph attention network was applied to prune the false-positive branches for reconstructing 3D liver vessel morphology, in which the GNN was only used as a post-processing without sufficient joint training and optimization with CNN. Shin *et al.* [28] proposed vessel graph network (VGN) to segment 2D blood vessels from color retinal images, in which a GNN was incorporated into a basic convolutional neural network (CNN) for 2D blood vessel segmentation. By VGN, the graphical features characterizing the vascular connectivity were generated by GNN and used to improve final predictions in the inference module of CNN. However, the GNN was embedded into CNN, and thus it would run with GNN in both training and inference stages. Consequently, VGN has two limitations: (i) The accuracy of connectivity in the graph cannot be well guaranteed because the graph is constructed using the intermediate predictions of CNN; (ii) The graph construction part cannot be computed in a parallel-based manner, which becomes a speed bottleneck of VGN. Particularly, this speed issue is significantly magnified in 3D images, making the time consumption of 3D VGN unacceptable in practical applications.

In this paper, we propose to segment 3D hepatic vessels from CT images by incorporating GNN-based connectivity constraints, which characterize the connectivity prior of hepatic vessels, into general CNN to improve the connectivity of segmented vessels. However, differing from [28], a new mechanism called the plug-in mode is applied to combine GNN and CNN, and GNN will be trained in a different way, so as to adapt to the 3D blood vessels segmentation and practical applications. First, an encoder-decoder deep convolutional neural network is proposed to perform voxel-wise segmentation, in which a lightweight design based on residual block [30] is applied to increase the patch size and reduce memory consumption while ensuring segmentation accuracy. Second, we directly use the ground truth of hepatic vessels to construct vascular connectivity graph which generates more accurate vascular connectivity than VGN [28]. As a result, an effective graph attention network (GAT) [31] can be used to model the graphical information of hepatic vessels. Third, based on the vascular connectivity graph and GAT, a graphical connectivity constraint module (GCCM) is designed, and incorporated into segmentation network as a multi-task branch. Here, the GCCM is integrated by a plug-in mode rather than embedding mode, which means that the GCCM can be used to improve vascular connectivity of segmentation network in the training stage, but will not run in the inference stage. Consequently, the proposed method not only can introduce more accurate connectivity constraints into 3D hepatic vessel segmentation, but also greatly reduce hardware and time costs of VGN. Extensive experiments are conducted on two public datasets, i.e., 3D-ircadb-01 [32] and MSD08 [33]. The results show that our method outperforms related methods, and can generate more accurate segmentation results and well improve vascular connectivity. The main contributions of this paper are summarized as follows:

1. A novel framework integrating CNN and GNN is proposed to segment 3D hepatic vessels for the first time, and thus the connectivity prior of hepatic vessels can be effectively utilized.
2. A new mechanism called a plug-in mode is applied to integrate GNN into CNN without increasing the hardware and time costs in the inference stage. Compared with the embedding mode in VGN, the proposed plug-in mode greatly reduces hardware and time costs in the inference stage and is more suitable for 3D images and clinical practice.
3. The vascular connectivity graph is constructed by ground truths and evaluated by geodesic distance, enhancing its accuracy of describing vascular connectivity prior.

Table

Dataset	3D-ircadb-01	MSD08
Original data	20	303
Original pixel spacing	0.57mm-0.87mm	0.57mm-0.98mm
Original slice thickness	1.0-4.0mm	0.8mm-8.0mm
Chosen data	20	89
Resampled pixel spacing	0.74mm	0.75mm
Resampled slice thickness	1.6mm	1.5mm
Cropped size	$192 \times 224 \times 80$	$384 \times 384 \times 96$
Limitation of CT values	[0,400] HU	
Normalization	Z-score normalization	

Conclusion

In this work, we propose a plug-in mechanism that integrates GNN into CNN to utilize connectivity prior to improve 3D hepatic vessel segmentation. The graphical connectivity information is constructed from ground truths and modeled by GAT, and the GAT is incorporated into the segmentation network as a multi-task branch, which supervises the training procedure of segmentation network with connectivity prior but will not run in the inference stage. Consequently, our proposed method effectively improves the accuracy and connectivity of hepatic vessel segmentation compared with related works, and greatly reduces hardware and time costs compared with VGN in [28]. The proposed method demonstrates a new way to introduce graphical connectivity information into deep neural networks, which can be further extended to other medical image analysis tasks.

2. **Continuous and Complete Liver Vessel Segmentation with Graph-Attention Guided Diffusion**

- Authors: Xiaotong Zhang, Alexander Broersen, Gonnie C. M. van Erp, Silvia L. Pintea, Jouke Dijkstra
- Year: 2024 · Source: arXiv
- URL: https://arxiv.org/abs/2411.00617
- Relevance: Recent SOTA on 3D-IRCADb-01 and LiVS arXivarXiv reporting that a graph-attention-guided diffusion model outperforms arXiv nnU-Net and Swin UNETR arXiv on connectivity/completeness ResearchGate +2 — an essential quantitative comparison target.

Abstract

Improving connectivity and completeness are the most challenging aspects of liver vessel segmentation, especially for small vessels. These challenges require both learning the continuous vessel geometry, and focusing on small vessel detection. However, current methods do not explicitly address these two aspects and cannot generalize well when constrained by inconsistent annotations. Here, we take advantage of the generalization of the diffusion model and explicitly integrate connectivity and completeness in our diffusion-based segmentation model. Specifically, we use a graph-attention module that adds knowledge about vessel geometry, and thus adds continuity. Additionally, we perform the graphattention at multiple-scales, thus focusing on small liver vessels. Our method outperforms eight state-of-the-art medical segmentation methods on two public datasets: 3D-ircadb-01 and LiVS. Our code is available at https://github.com/ZhangXiaotong015/GATSegDiff.

Table

	Type	*DSC* (%)	*clDice* (%)	*Sen* (%)	*Spe* (%)	*Con* ($\to 1$)
Gao et al.[24]	2D	60.19 ± 4.69	56.48 ± 11.74	64.98 ± 4.48	99.80 ± 0.10	8.73 (96/11)
nnUNet[14]	3D	60.54 ± 6.86	71.60 ± 4.37	44.76 ± 8.09	$\mathbf{99.99} \pm 0.00$	2.36 (26/11)
Swin UNETR[15]	3D	55.61 ± 9.80	62.75 ± 7.91	42.30 ± 11.46	99.98 ± 0.01	5.45 (60/11)
TransUNet[55]	3D	69.77 ± 3.73	$\mathbf{75.83} \pm 5.50$	58.63 ± 11.70	99.97 ± 0.01	3.73 (41/11)
EnsemDiff[38]	2.5D	55.07 ± 9.70	60.99 ± 9.64	40.45 ± 10.60	99.98 ± 0.02	7.18 (79/11)
MedSegDiff[39]	2.5D	59.67 ± 7.74	66.02 ± 7.87	47.42 ± 10.43	99.95 ± 0.05	8.64 (95/11)
MERIT[54]	2.5D	49.50 ± 9.57	53.00 ± 8.79	34.30 ± 8.81	99.99 ± 0.01	5.50 (59/11)
HiDiff[42]	2.5D	63.32 ± 4.01	60.24 ± 2.71	54.45 ± 7.25	99.94 ± 0.01	6.70 (72/11)
Ours	2.5D	$\mathbf{71.26} \pm 1.93$	74.61 ± 1.21	$\mathbf{71.59} \pm 4.07$	99.89 ± 0.04	**1.09** (12/11)

Conclusion

In this study, we focus on liver vessel segmentation from CT volumes. To this end, we propose to augment conditional diffusion models with geometric graph-structure computed at multiple resolutions. The role of the graph structure is to ensure connectivity in the segmentation, across neighboring slices in the CT volume. Moreover, we use multi-scale features in the graph, thus allowing the model to focus on small vessels, that otherwise would be missed. Our proposed model achieves state-of-the-art results, in terms of DSC, Sen and vessel connectivity on two standard benchmarks: 3D-ircadb-01 [23] and LiVS [24], when compared to baselines such as HiDiff [42], MERIT[54], TransUNet [55], 16 MedSegDiff [39], EnsemDiff [38], Swin UNETR [15], nnUNet [14] and Gao et al. [24].

3. **Scale-specific Auxiliary Multi-task Contrastive Learning for Deep Liver Vessel Segmentation**

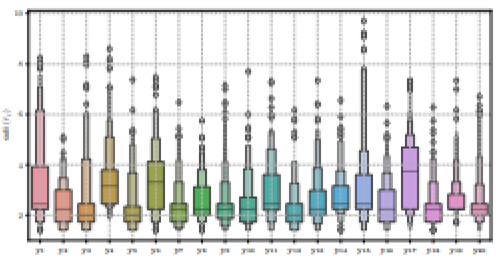
- Authors: Amine Sadikine, Bogdan Badic, Jean-Pierre Tasu, Vincent Noblet, Pascal Ballet, Dimitris Visvikis, Pierre-Henri Conze
- Year: 2024 · Source: arXiv (ISBI 2024)
- URL: https://arxiv.org/abs/2409.12333
- Relevance: Adds scale-specific auxiliary heads and contrastive learning arXiv to a 3D U-Net to preserve thin peripheral branches arXiv — addresses exactly the vessel-calibre imbalance that clDice targets.

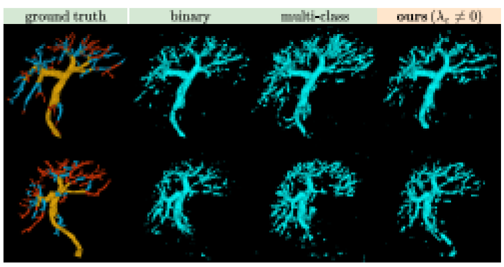
Abstract

Extracting hepatic vessels from abdominal images is of high interest for clinicians since it allows to divide the liver into functionally-independent Couinaud segments. In this respect, an automated liver blood vessel extraction is widely summoned. Despite the significant growth in performance of semantic segmentation methodologies, preserving the complex multi-scale geometry of main vessels and ramifications remains a major challenge. This paper provides a new deep supervised approach for vessel segmentation, with a strong focus on representations arising from the different scales inherent to the vascular tree geometry. In particular, we propose a new clustering technique to decompose the tree into various scale levels, from tiny to large vessels. Then, we extend standard 3D UNet to multi-task learning by incorporating scalespecific auxiliary tasks and contrastive learning to encourage the discrimination between scales in the shared representation. Promising results, depicted in several evaluation metrics, are revealed on the public 3D-IRCADb dataset.

Table

Models		DSC↑ score (%)	Jacc↑ score (%)	clDSC↑ score (%)	HD↓ dist. (mm)
3D ResUNet	binary	54.25 ± 3.65	37.65 ± 3.46	46.84 ± 3.63	**80.09** ± 4.83
	multi-class	50.31 ± 4.43	34.13 ± 3.98	44.34 ± 2.89	114.3 ± 10.9
Ours	$\lambda_c = 0$	54.27 ± 4.16	37.73 ± 3.87	46.90 ± 2.44	86.07 ± 8.07
	$\lambda_c \neq 0$	**55.04** ± 2.79	**38.50** ± 2.71	**48.13** ± 3.52	82.36 ± 12.3





Conclusion

We have unveiled a novel approach for liver vessel segmentation based on scale-specific auxiliary multi-task contrastive learning. In future works, it would be valuable to evaluate the proposed method with a various number of vascular tree scales to more deeply assess its robustness. In addition, incorporating shape and topological priors could allow the model to avoid prediction confusion and increase its generalization ability. Furthermore, a memory bank storing multi-scale latent representations could be a way to further benefit from contrastive learning while alleviating batch size constraints. More globally, our contributions could be easily extended to other types of vasculature including brain and pulmonary vessels from diverse imaging modalities.

4. **Deep Vessel Segmentation with Joint Multi-Prior Encoding**

- Authors: Amine Sadikine, Bogdan Badic, Jean-Pierre Tasu, Vincent Noblet, Dimitris Visvikis, Pierre-Henri Conze
- Year: 2024 · Source: arXiv (ISBI 2024)
- URL: https://arxiv.org/abs/2409.12334
- Relevance: Jointly encodes shape and topological priors arXiv in a shared encoder for liver vessels on 3D-IRCADb arXiv — a complementary alternative to clDice for topology awareness.

Abstract

The precise delineation of blood vessels in medical images is critical for many clinical applications, including pathology detection and surgical planning. However, fully-automated vascular segmentation is challenging because of the variability in shape, size, and topology. Manual segmentation remains the gold standard but is time-consuming, subjective, and impractical for large-scale studies. Hence, there is a need for automatic and reliable segmentation methods that can accurately detect blood vessels from medical images. The integration of shape and topological priors into vessel segmentation models has been shown to improve segmentation accuracy by offering contextual information about the shape of the blood vessels and their spatial relationships within the vascular tree. To further improve anatomical consistency, we propose a new joint prior encoding mechanism which incorporates both shape and topology in a single latent space. The effectiveness of our method is demonstrated on the publicly available 3D-IRCADb dataset. More globally, the proposed approach holds promise in overcoming the challenges associated with automatic vessel delineation and has the potential to advance the field of deep priors encoding.

Table

Models		DSC↑ score (%)	Jacc↑ score (%)	clDSC↑ score (%)	HD↓ dist. (mm)	AVD↓ dist. (mm^3)	ASSD↓ dist. (mm)
ResUNet	-	54.06 ± 2.34	37.31 ± 2.14	46.31 ± 2.37	70.88 ± 9.20	0.36 ± 0.18	5.23 ± 0.75
ResUNet+shape	$\lambda_s = 26.21$	53.50 ± 2.44	36.97 ± 2.31	44.50 ± 3.03	$\underline{61.88} \pm 5.02$	0.35 ± 0.18	5.20 ± 0.69
ResUNet+topo	$\lambda_t = 32.01$	53.50 ± 1.50	36.84 ± 1.40	47.41 ± 1.81	$\mathbf{61.72} \pm 5.74$	0.37 ± 0.15	5.09 ± 0.73
ResUNet+shape+topo	$\lambda_s = 63.33, \lambda_t = 14.53$	$\underline{54.59} \pm 2.06$	$\underline{37.88} \pm 1.87$	$\underline{47.63} \pm 2.72$	71.50 ± 8.44	$\mathbf{0.33} \pm 0.15$	$\underline{4.81} \pm 0.72$
Ours	$\lambda = 65.10$	$\mathbf{54.78} \pm 2.35$	$\mathbf{38.00} \pm 2.17$	$\mathbf{50.34} \pm 2.84$	67.06 ± 5.27	$\underline{0.34} \pm 0.15$	$\mathbf{4.77} \pm 0.74$

Results and Discussion

for liver vessel segmentation, assessed using various evaluation metrics including DSC (Dice Similarity Coefficient), Jacc (Jaccard score), clDSC coefficient [17] for connectivity assessment, HD (Hausdorff distance), AVD (Absolute Volume Difference), and ASSD (Average Symmetric Surface Distance). The models include ResUNet as baseline, ResUNet+shape, ResUNet+topo, ResUNet+shape+topo, and the proposed approch. Our method outperforms the other models, achieving the highest DSC, Jacc, clDSC, and ASSD scores with 54.78%, 38.00%, 50.34%, and 4.77mm respectively. It delivers robust performance in clDSC assessment, positioning it as a promising topology-aware model. Further, Fig.3 illustrates the connectivity improvement reached by our approach. In particular, fine branches remain less disconnected from main vessels compared to ResUNet+topo or ResUNet+shape+topo. The performance of our approach could be improved by calibrating the hyper-parameter α p (Eq.10), which indirectly affects the JMPE coding scheme. Furthermore, our method allows the use of a single encoder instead of multiple encoders, thus reducing memory consumption.

Conclusion

In this paper, we presented an innovative approach that addresses the integration of multiple priors into a unified formulation for segmentation purposes. The liver vessel delineation results obtained from our method highlight the importance of incorporating high-level and topological constraints in medical image segmentation, and provide potential avenues for future research in this area. Furthermore, extending this approach to other datasets could provide valuable insights into its generalizability and effectiveness in various clinical contexts. Additionally, integrating graph neural networks in our pipeline could further enhance connectivity encoding.

5. **Deep Vessel Segmentation Based on a New Combination of Vesselness Filters**

- Authors: Guillaume Garret, Antoine Vacavant, Carole Frindel
- Year: 2024 · Source: arXiv
- URL: https://arxiv.org/abs/2402.14509
- Relevance: Studies Frangi/Jerman/Sato vesselness maps as auxiliary inputs to a U-Net for liver CTA and brain MRA arXiv — useful preprocessing baseline against nnU-Net v2's normalisation-only pipeline.

Abstract

Vascular segmentation represents a crucial clinical task, yet its automation remains challenging. Because of the recent strides in deep learning, vesselness filters, which can significantly aid the learning process, have been overlooked. This study introduces an innovative filter fusion method crafted to amplify the effectiveness of vessel segmentation models. Our investigation seeks to establish the merits of a filter-based learning approach through a comparative analysis. Specifically, we contrast the performance of a U-Net model trained on CT images with an identical U-Net configuration trained on vesselness hyper-volumes using matching parameters. Our findings, based on two vascular datasets, highlight improved segmentations, especially for small vessels, when the model's learning is exposed to vessel-enhanced inputs.

Results

Our analysis on the IRCAD dataset, detailed in Table 2(a), reveals a notable improvement in the mean Dice for small vessels with the application of enhancement filters, with an average deviation over +9%. However, a negative impact on large vessels is observed, likely due to signal loss in our enhanced dataset (see Fig. 2). Adjusting the scale space of filters may mitigate this. Bifurcation segmentation slightly decreased in the test set, traced back to a specific sample's drop in Dice score under our vesselness configuration. Additionally, vessel-enhancement improved the preservation of vascular topology, reflected in increased clDice scores across partitions. On test sets, improvements ranged from an average deviation of +1.47% (medium vessels) to +16% (small vessels), and on validation sets, from +4.16% to +19.7%. Visual representation is provided in Fig. 3.

For the Bullitt dataset, we have improved segmentation with regard to our both metrics, as shown in Table 2(b). This is consistent with the previous results, as the Bullitt's vessels are significantly smaller than those of IRCAD. Hence we are not impacted by large vessels signal loss. In addition, we note that the vesselness benefit is less for

Bullitt, from +1.09% to +3.88%, for bifurcations and medium vessels respectively. This smaller average deviations can be explained by the fact that the Bullitt dataset is almost noise-free compared to the IRCAD, which reduces the impact of the filters. Indeed, the Peak Signal-to-Noise Ratio (PSNR) for the original Bullitt of 19.04 ± 0.43 improved to just 21.56 ± 1.12 after enhancement, compared to IRCAD, which increase significantly from 9.35 ± 1, 22 to 19.74 ± 1.60 after enhancement. An illustration of segmentation is given Figure 3.

Table

2(a) IRCAD

Dice	Original		Vessel enhanced	
	Test	Validation	Test	Validation
Global	**0.631 ± 0.044**	**0.645 ± 0.021**	**0.638 ± 0.071**	**0.672 ± 0.021**
Large	**0.745 ± 0.094**	**0.775 ± 0.015**	**0.708 ± 0.082**	**0.667 ± 0.018**
Medium	**0.645 ± 0.086**	**0.629 ± 0.059**	**0.668 ± 0.092**	**0.695 ± 0.032**
Small	**0.430 ± 0.054**	**0.389 ± 0.053**	**0.522 ± 0.04**	**0.515 ± 0.029**
Bifurcations	**0.728 ± 0.139**	**0.809 ± 0.036**	**0.718 ± 0.124**	**0.853 ± 0.041**

clDice	Original		Vessel enhanced	
	Test	Validation	Test	Validation
Global	**0.614 ± 0.039**	**0.623 ± 0.061**	**0.720 ± 0.104**	**0.763 ± 0.027**
Large	**0.828 ± 0.059**	**0.822 ± 0.028**	**0.860 ± 0.049**	**0.864 ± 0.025**
Medium	**0.733 ± 0.086**	**0.751 ± 0.077**	**0.748 ± 0.165**	**0.852 ± 0.029**
Small	**0.483 ± 0.071**	**0.458 ± 0.087**	**0.645 ± 0.051**	**0.655 ± 0.037**
Bifurcations	**0.797 ± 0.067**	**0.805 ± 0.061**	**0.866 ± 0.054**	**0.899 ± 0.014**

2(b) Bullitt

Dice	Original	Vessel enhanced
Global	0.642 ± 0.004	**0.656 ± 0.002**
Medium	0.709 ± 0.012	**0.748 ± 0.008**
Small	0.685 ± 0.005	**0.699 ± 0.003**
Bifurcations	0.888 ± 0.005	**0.899 ± 0.004**

clDice	Original	Vessel enhanced
Global	0.775 ± 0.005	**0.783 ± 0.002**
Medium	0.760 ± 0.014	**0.791 ± 0.008**
Small	0.831 ± 0.006	**0.844 ± 0.004**
Bifurcations	0.935 ± 0.003	**0.942 ± 0.003**

Conclusion

In this study, we introduced a novel fusion method for vesselness filters in filter-based learning, concatenating filters into a hyper-volume. Our comparative study, utilizing a U-Net model with fixed parameters, assessed the model's efficiency when trained on original images versus our proposed hyper-volumes. The results yielded valuable insights into the efficacy of our approach. Notably, the incorporation of vesselness-based learning demonstrated an improvement in the Dice score, particularly for small vessels, and preserved vascular structure topology, as indicated by the clDice measure. We evaluated our approach on two diverse datasets—Bullitt for cerebral MR and IRCAD for hepatic CT applications—following a partitioning approach. Looking ahead, our future work will delve into examining the contribution of the studied filters using eXplainable AI methods. We believe this approach could reveal the most influential filters for the models, enhancing our understanding of the model's behavior.

6. **Hepatic Vessel Segmentation Based on 3D Swin-Transformer with Inductive-Biased Multi-Head Self-Attention (IBIMHAV-Net)**

- Authors: Mian Wu, Yinling Qian, Xiangyun Liao, Qiong Wang, Pheng-Ann Heng
- Year: 2021 · Source: arXiv
- URL: https://arxiv.org/abs/2111.03368
- Relevance: Task-specific 3D Swin-Transformer for hepatic vessels ResearchGate on 3DIRCADb PubMed CentralarXiv — strong transformer baseline to benchmark against the nnU-Net v2 multiplanar CNN.

Abstract

Purpose: Segmentation of liver vessels from CT images is indispensable prior to surgical planning and aroused broad range of interests in the medical image analysis community. Due to the complex structure and low contrast background, automatic liver vessel segmentation remains particularly challenging. Most of the related researches adopt FCN, U-net, and V-net variants as a backbone. However, these methods mainly focus on capturing multi-scale local features which may produce misclassified voxels due to the convolutional operator's limited locality reception field.

Methods:

We propose a robust end-to-end vessel segmentation network called Inductive BIased Multi-Head Attention Vessel Net(IBIMHAV-Net) by expanding swin transformer to 3D and employing an effective combination of convolution and self-attention. In practice, we introduce the voxel-wise embedding rather than patch-wise embedding to locate precise liver vessel voxels, and adopt multi-scale convolutional operators to gain local spatial information. On the other hand, we propose the inductive biased multi-head self-attention which learns inductive biased relative positional embedding from initialized absolute position embedding. Based on this, we can gain a more reliable query and key matrix. To validate the generalization of our model, we test on samples which have different structural complexity.

Results:

We conducted experiments on the 3DIRCADb datasets. The average dice and sensitivity of the four tested cases were 74.8% and 77.5%, which exceed results of existing deep learning methods and improved graph cuts method.

Figure

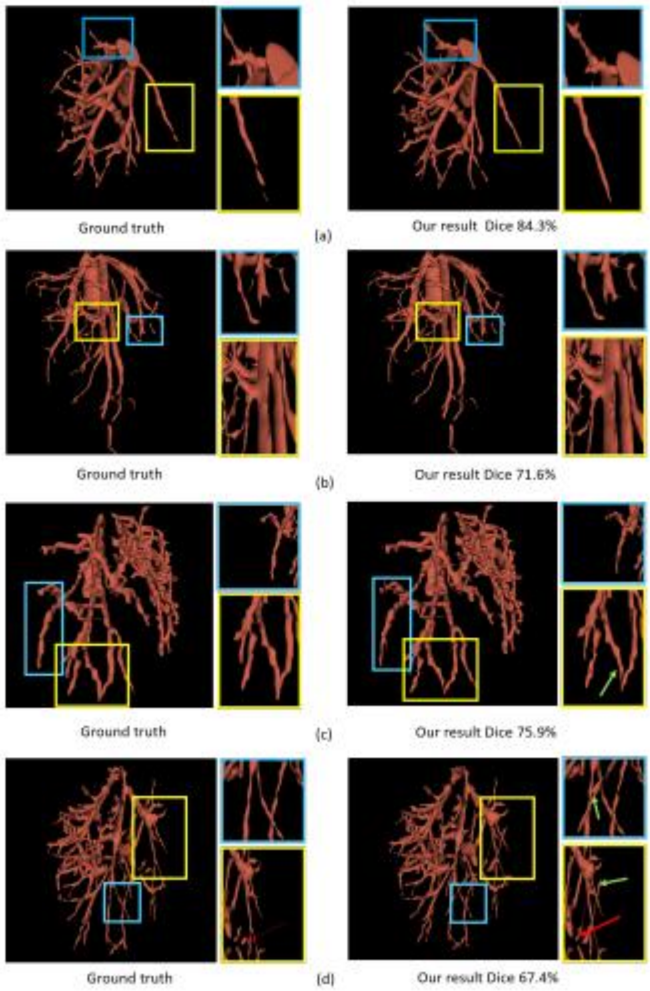


Fig. 5 The first column list ground truth in different cases. The second column list our network's results (a),(b),(c),(d) represent different cases.

Conclusion

This paper designs a liver vessels segmentation method from CT images using the transformer-based network. Swin transformer has been expanding to 3D as the backbone which interleaved with convolutions and expanding for 3D volumes. In specific, the small stride convolution in both local feature block path and up/down-sampling blocks keep the spatial information hierarchically for two successive swin transformer blocks. A new pixel-wised embedding method has been used for our few samples task with variant structures. A new type of bias positional embedding in our transformer is proposed. Numerical Evaluation and visualization based on different benchmarks proved the validity of this deep learning method. Our method has been trained and tested on 3DIRCADb datasets. In the future, we would further improve segmentation accuracy by introducing more precise datesets and trying multi-task method to reduce negative effects of liver tumors.

7. **Enhancing the Automatic Segmentation and Analysis of 3D Liver Vasculature Models**

- Authors: Y. Nasser et al.
- Year: 2024 · Source: arXiv
- URL: https://arxiv.org/abs/2411.15778
- Relevance: **Direct precedent —** benchmarks clDice and skeleton-based losses on portal/hepatic venous trees and uses connected-component and skeleton post-processing to produce multi-class outputs, matching this project's exact combination.

Abstract

. Surgical assessment of liver cancer patients requires identification of the vessel trees from medical images. Specifically, the venous trees - the portal (perfusing) and the hepatic (draining) trees are important for understanding the liver anatomy and disease state, and perform surgery planning. This research aims to improve the 3D segmentation, skeletonization, and subsequent analysis of vessel trees, by creating an automatic pipeline based on deep learning and image processing techniques. The first part of this work explores the impact of differentiable skeletonization methods such as ClDice and morphological skeletonization loss, on the overall liver vessel segmentation performance. To this aim, it studies how to improve vessel tree connectivity. The second part of this study converts a single class vessel segmentation into multi-class ones, separating the two venous trees. It builds on the previous two-class vessel segmentation model, which vessel tree outputs might be entangled, and on connected components and skeleton analyses of the trees. After providing sub-labeling of the specific anatomical branches of each venous tree, these algorithms also enable a morphometric analysis of the vessel trees by extracting various geometrical markers. In conclusion, we propose a method that successfully improves current skeletonization methods, for extensive vascular trees that contain vessels of different calibers. The separation algorithm creates a clean multi-class segmentation of the vessels, validated by surgeons to provide low error. A new, publicly shared high-quality liver vessel dataset of 77 cases is thus created. Finally a method to annotate vessel trees according to anatomy is provided, enabling a unique liver vessel morphometry analysis.

Discussion

The results show that the differentiable skeletonization performance was lackluster compared to the widely used but non-differentiable version [17]. The skeletonization neural network achieves much better centerline representations than the morphological operations as shown in Table 1 and Fig.3. Moreover, we investigated whether skeleton quality might improve ClDice results. Adding a neural network inference instead of a skeletonization algorithm consisting of a series of morphological operations does double the training time (from 80s to 160s per epoch) but it provides a more accurate and homogeneous skeleton for the ClDice computation. However, the overall segmentation performance does not improve as shown in Table 2. This result holds even when training for a larger number of epochs. The pretrained aspect of this neural network can make it multipurpose and is its biggest advantage. Since its input is a segmentation volume, it can theoretically be a general skeletonization algorithm for any tubular structure (such as other organs' vessels or tree roots for example).

Conclusion

In this paper we showcase a way to train a differentiable and precise skeletonization algorithm, a deep dive of ClDice on liver vessels which feature a 3D structure, with an extensive network of vessels that have vastly different radii. We also present a method to obtain multi-label segmentations. This makes the annotation burden lighter as we do not need to pay attention to which tree a vessel tree belongs to, and automatically label the trees after. Finally we also introduce a novel method to automatically label the main clinically relevant branches for surgeons during their assessment, such as the main branches and their immediate daughter branches. We also propose a labeling of the smaller branches according to their generation or Couinaud region. Finally, we extract the first computational morphometric study on human liver vessels as far as we are aware.

Multi-planar, multi-view, and 2.5D approaches

8. **DeepOrgan: Multi-level Deep Convolutional Networks for Automated Pancreas Segmentation**

- Authors: Holger R. Roth, Le Lu, Amal Farag, Hoo-Chang Shin, Jiamin Liu, Evrim Turkbey, Ronald M. Summers
- Year: 2015 (MICCAI) · Source: arXiv
- URL: https://arxiv.org/abs/1506.06448
- Relevance: Early canonical 2.5D tri-planar CNN using orthogonal axial/coronal/sagittal patches fused probabilistically arXiv — conceptual ancestor of the project's pipeline.

Abstract

Automatic organ segmentation is an important yet challenging problem for medical image analysis. The pancreas is an abdominal organ with very high anatomical variability. This inhibits previous segmentation methods from achieving high accuracies, especially compared to other organs such as the liver, heart or kidneys. In this paper, we present a probabilistic bottom-up approach for pancreas segmentation in abdominal computed tomography (CT) scans, using multi-level deep convolutional networks (ConvNets). We propose and evaluate several variations of deep ConvNets in the context of hierarchical, coarse-to-fine classification on image patches and regions, i.e. superpixels. We first present a dense labeling of local image patches via P-ConvNet and nearest neighbor fusion. Then we describe a regional ConvNet (R1−ConvNet) that samples a set of bounding boxes around each image superpixel at different scales of contexts in a “zoom-out” fashion. Our ConvNets learn to assign class probabilities for each superpixel region of being pancreas. Last, we study a stacked R2−ConvNet leveraging the joint space of CT intensities and the P−ConvNet dense probability maps. Both 3D Gaussian smoothing and 2D conditional random fields are exploited as structured predictions for post-processing. We evaluate on CT images of 82 patients in 4-fold crossvalidation. We achieve a Dice Similarity Coefficient of 83.6±6.3% in training and 71.8±10.7% in testing.

Introduction

Segmentation of the pancreas can be a prerequisite for computer aided diagnosis (CADx) systems that provide quantitative organ volume analysis, e.g. for diabetic patients. Accurate segmentation could also necessary for computer aided detection (CADe) methods to detect pancreatic cancer. Automatic segmentation of numerous organs in computed tomography (CT) scans is well studied with good performance for organs such as liver, heart or kidneys, where Dice Similarity Coefficients (DSC) of >90% are typically achieved Wang et al. (2014), Chu et al. (2013), Wolz et al. (2013), Ling et al.

(2008). However, achieving high accuracies in automatic pancreas segmentation is still a challenging task. The pancreas' shape, size and location in the abdomen can vary drastically between patients. Visceral fat around the pancreas can cause large variations in contrast along its boundaries in CT (see Fig. 3). Previous methods report only 46.6% to 69.1% DSCs Wang et al. (2014), Chu et al. (2013), Wolz et al. (2013), Farag et al. (2014). Recently, the availability of large annotated datasets and the accessibility of affordable parallel computing resources via GPUs have made it feasible to train deep convolutional networks (ConvNets) for image classification. Great advances in natural image classification have been achieved Krizhevsky et al. (2012). However, deep ConvNets for semantic image segmentation have not been well studied Mostajabi et al. (2014). Studies that applied ConvNets to medical imaging applications also show good promise on detection tasks Cire ¸san et al. (2013), Roth et al. (2014). In this paper, we extend and exploit ConvNets for a challenging organ segmentation problem.

Methods

We present a coarse-to-fine classification scheme with progressive pruning for pancreas segmentation. Compared with previous top-down multi-atlas registration and label fusion methods, our models approach the problem in a bottom-up fashion: from dense labeling of image patches, to regions, and the entire organ. Given an input abdomen CT, an initial set of superpixel regions is generated by a coarse cascade process of random forests based pancreas segmentation as proposed by Farag et al. (2014). These pre-segmented superpixels serve as regional candidates with high sensitivity (>97%) but low precision. The resulting initial DSC is ~27% on average. Next, we propose and evaluate several variations of ConvNets for segmentation refinement (or pruning). A dense local image patch labeling using an axial-coronal-sagittal viewed patch (P−ConvNet) is employed in a sliding window manner. This generates a perlocation probability response map P. A regional ConvNet (R1−ConvNet) samples a set of bounding boxes covering each image superpixel at multiple spatial scales in a "zoom-out" fashion Mostajabi et al. (2014), Girshick et al. (2014) and assigns probabilities of being pancreatic tissue. This means that we not only look at the close-up view of superpixels, but gradually add more contexts to each candidate region. R1-ConvNet operates directly on the CT intensity. Finally, a stacked regional R2−ConvNet is learned to leverage the joint convolutional features of CT intensities and probability maps P. Both 3D Gaussian smoothing and 2D conditional random fields for structured prediction are exploited as post-processing. Our methods are evaluated on CT scans of 82 patients in 4-fold cross-validation (rather than "leaveone-out" evaluation Wang et al. (2014), Chu et al. (2013), Wolz et al. (2013)). We propose several new ConvNet models and advance the current state-of-the-art performance to a DSC of 71.8 in testing. To the best of our knowledge, this is the highest DSC reported in the literature to date.

Table

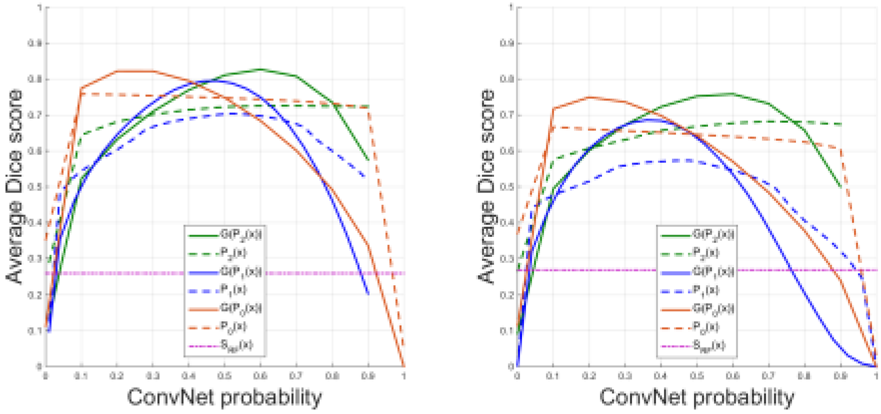


Figure 5: Average DSCs as a function of un-smoothed $P_k(x), k = 0, 1, 2$, and 3D smoothed $G(P_k(x)), k = 0, 1, 2$, ConvNet probability maps in training (left) and testing (right) in one cross-validation fold

Table 1: **4-fold cross-validation**: optimally achievable DSCs, our initial candidate region labeling using S_{RF}, DSCs on $P(x)$ and using smoothed $G(P(x))$, and a CRF model for structured prediction (best performance in bold).

DSC (%)	Opt.	$S_{RF(x)}$	$P_0(x)$	$G(P_0(x))$	$P_1(x)$	$G(P_1(x))$	$P_2(x)$	$G(P_2(x))$	$CRF(P_2(x))$
Mean	80.5	26.1	60.9	69.5	56.8	62.9	64.9	**71.8**	68.2
Std	3.6	7.1	10.4	9.3	11.4	16.1	8.1	**10.7**	4.1
Min	70.9	14.2	22.9	35.3	1.3	0.0	33.1	**25.0**	59.6
Max	85.9	45.8	80.1	84.4	77.4	87.3	77.9	**86.9**	74.2

Conclusion

We present a bottom-up, coarse-to-fine approach for pancreas segmentation in abdominal CT scans. Multi-level deep ConvNets are employed on both image patches and regions. We achieve the highest reported DSCs of 71.8±10.7% in testing and 83.6±6.3% in training, at the computational cost of a few minutes, not hours as in Wang et al. (2014), Chu et al. (2013), Wolz et al. (2013). The proposed approach can be incorporated into multi-organ segmentation frameworks by specifying more tissue types since ConvNet naturally supports multi-class classifications Krizhevsky et al. (2012). Our deep learning based organ segmentation approach could be generalizable to other segmentation problems with large variations and pathologies, e.g. tumors.

9. **Multi-Planar Deep Segmentation Networks for Cardiac Substructures from MRI and CT**

- Authors: Aliasghar Mortazi, Jeremy Burt, Ulas Bagci
- Year: 2017 · Source: arXiv
- URL: https://arxiv.org/abs/1708.00983
- Relevance: Trains three plane-wise 2D CNNs with adaptive probability-map fusion arXivPubMed Central — the canonical architectural analogue of the proposed axial/sagittal/coronal nnU-Net pipeline.

Abstract

Non-invasive detection of cardiovascular disorders from radiology scans requires quantitative image analysis of the heart and its substructures. There are well-established measurements that radiologists use for diseases assessment such as ejection fraction, volume of four chambers, and myocardium mass. These measurements are derived as outcomes of precise segmentation of the heart and its substructures. The aim of this paper is to provide such measurements through an accurate image segmentation algorithm that automatically delineates seven substructures of the heart from MRI and/or CT scans. Our proposed method is based on multi-planar deep convolutional neural networks (CNN) with an adaptive fusion strategy where we automatically utilize complementary information from different planes of the 3D scans for improved delineations. For CT and MRI, we have separately designed three CNNs (the same architectural configuration) for three planes, and have trained the networks from scratch for voxel-wise labeling for the following cardiac structures: myocardium of left ventricle (Myo), left atrium (LA), left ventricle (LV), right atrium (RA), right ventricle (RV), ascending aorta (Ao), and main pulmonary artery (PA). We have evaluated the proposed method with 4-fold-cross-validation on the multi-modality whole heart segmentation challenge (MM-WHS 2017) dataset. The precision and dice index of 0.93 and 0.90, and 0.87 and 0.85 were achieved for CT and MR images, respectively. While a CT volume was segmented about 50 seconds, an MRI scan was segmented around 17 seconds with the GPUs/CUDA implementation.

Table

	MRI								CT							
Structures:	*Myo*	*LA*	*LV*	*RA*	*RV*	*Aorta*	*PA*	*WHS*	*Myo*	*LA*	*LV*	*RA*	*RV*	*Aorta*	*PA*	*WHS*
Sensitivity	0.816	0.856	0.928	0.827	0.878	0.728	0.782	0.831	0.888	0.903	0.918	0.835	0.872	0.86	0.783	0.866
Specificity	0.999	0.999	0.999	0.998	0.999	0.999	0.998	0.999	0.999	0.999	0.999	0.999	0.998	0.999	0.999	0.999
Precision	0.842	0.88	0.936	0.909	0.846	0.873	0.791	0.868	0.912	0.931	0.944	0.914	0.911	0.983	0.912	0.929
DI	0.825	0.887	0.932	0.874	0.884	0.772	0.784	0.851	0.898	0.925	0.93	0.877	0.888	0.909	0.851	0.897
S2S(mm)	1.152	1.130	1.084	1.401	1.825	1.977	2.287	1.551	0.903	1.386	1.142	2.019	1.896	1.023	1.781	1.450

Discussion and Conclusion

The main goal of the current study is to develop a framework for accurately segmenting the all cardiac substructures from both CT and MR images with high efficiency. The main strength of the proposed method is to train multiple CNNs from scratch and to allow an adaptive fusion strategy for information maximization in pixel labeling despite the limited data and hardware support. Our findings indicate that MO-MP-CNN can be used as an efficient tool to delineate cardiac structures with high precision, accuracy, and efficiency. Technically, one may question why we did not employ a completely 3D CNN approach instead of utilizing a multi-planar fusion of multiple 2D CNNs. As discussed in [5], the lack of a large number of 3D images restricts the depth of CNN training, which may highly likely result in sub-optimal implementation. Hence, training large number of 2D slices is much more feasible than utilizing 3D approach with the current setting. In the instance of plentiful GPU processing power and 3D imaging data, training would be optimized using a 3D CNN. Another limitation of our work stems from the use of the softmax function in the last layer of the proposed network. To explore whether the information loss due to class normalization in this step is significant, further research should be undertaken using information from the layer before the softmax in fusion part and compared with the current system. Finally, further work is needed to establish comparative evaluation of different deep neural network approaches such as ResNet, U-net, and others. While deeper networks are desirable to achieve higher precision in segmentation tasks, lack of 3D data is a significant limitation for training such a system. Data augmentation and transfer learning have been shown to adequately address such challenges to a certain degree, but there is currently no research proving the optimality of such networks relative to the availability of data at hand.

10. **One Network to Segment Them All: A General, Lightweight System for Accurate 3D Medical Image Segmentation (Multi-Planar U-Net)**

- Authors: Mathias Perslev, Erik B. Dam, Akshay Pai, Christian Igel
- Year: 2019 (MICCAI) · Source: arXiv
- URL: https://arxiv.org/abs/1911.01764
- Relevance: MPUnet samples 2D slices across multiple 3D axes and fuses per-view predictions with learned weights Wiley Online LibraryGitHub — an MSD-2018 top performer and obvious head-to-head baseline.

Abstract

Many recent medical segmentation systems rely on powerful deep learning models to solve highly specific tasks. To maximize performance, it is standard practice to evaluate numerous pipelines with varying model topologies, optimization parameters, pre- & postprocessing steps, and even model cascades. It is often not clear how the resulting pipeline transfers to different tasks. We propose a simple and thoroughly evaluated deep learning framework for segmentation of arbitrary medical image volumes. The system requires no task-specific information, no human interaction and is based on a fixed model topology and a fixed hyperparameter set, eliminating the process of model selection and its inherent tendency to cause methodlevel over-fitting. The system is available in open source and does not require deep learning expertise to use. Without task-specific modifications, the system performed better than or similar to highly specialized deep learning methods across 3 separate segmentation tasks. In addition, it ranked 5-th and 6-th in the first and second round of the 2018 Medical Segmentation Decathlon comprising another 10 tasks. The system relies on multi-planar data augmentation which facilitates the application of a single 2D architecture based on the familiar U-Net. Multi-planar training combines the parameter efficiency of a 2D fully convolutional neural network with a systematic train- and test-time augmentation scheme, which allows the 2D model to learn a representation of the 3D image volume that fosters generalization.

Table

Table 1. Performance of the MPUnet across thirteen segmentation tasks. The shown F1 (dice) scores are mean values computed across all non-background per-class F1 scores. For the 10 MSD datasets evaluation was performed by the challenge organisers on non-publicly available test-sets. For MICCAI and HarP, evaluation was performed over three trials. Five fold cross-validation was used for OAI. The 'Classes' column include the background class, which is not included when computing the F1 scores. The 'Size' column gives the total dataset size. Note that the F1 standard deviations for tasks 8, 9 & 10 are not yet published by the challenge organizers. We refer to `http://medicaldecathlon.com/results.html` for a detailed comparison of our results (team CerebriuDIKU) with those of other challenge participants.

	Dataset	Modality	Segmentation Target(s)	Classes	Size	F1 Score
	MICCAI	MRI	Whole-Brain	135	35	0.74 ± 0.03
	HarP	MRI	L+R Hippocampus	3	135	0.85 ± 0.03
	OAI	MRI	Knee Cartilages	7	176	0.87 ± 0.06
2018 Medical Segmentation Decathlon	Task 1	MRI	Brain Tumours	4	750	0.60 ± 0.24
	Task 2	MRI	Cardiac, Left Atrium	2	30	0.89 ± 0.09
	Task 3	CT	Liver & Tumour	2	201	0.76 ± 0.18
	Task 4	MRI	Hippocampus ROI.	2	394	0.89 ± 0.04
	Task 5	MRI	Prostate	3	48	0.78 ± 0.10
	Task 6	CT	Lung Tumours	2	96	0.59 ± 0.23
	Task 7	CT	Pancreas & Tumour	3	420	0.48 ± 0.21
	Task 8	CT	Hepatic Ves. & Tumour	3	443	0.49
	Task 9	CT	Spleen	2	61	0.95
	Task 10	CT	Colon Cancer	2	190	0.28

Discussion and Conclusion

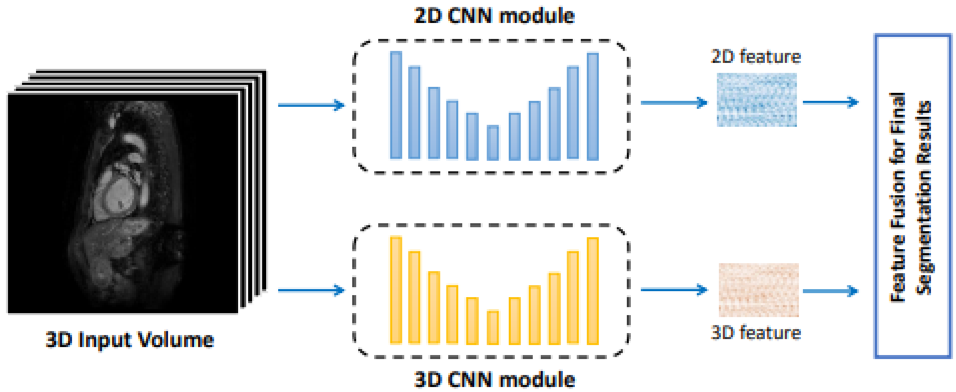
The empirical evaluation over 13 segmentation tasks showed that multi-planar augmentation provides a simple mechanism for obtaining accurate segmentation models without hyperparameter tuning. With no task-specific modifications the MPUnet performs well across many non-pathological tissues imaged with various MR and CT protocols, in spite of the target compartments varying drastically in number, physical size, shape- and spatial distributions, as well as contrast to the surrounding tissues. Also the accuracies on the more difficult pathological targets are favorable compared to most other MSD contesters. The MSD winning algorithm [14] relied on selecting a suitable model topology and/or cascade from an ensemble of candidates through cross-validation. In contrast to this and other top-ranking participants, we were interested to develop a task-agnostic segmentation system based on a single architecture and learning procedure that makes the system lightweight and easily transferable to clinical settings with limited compute resources. That the MPUnet can be applied 'as is' across many tasks with high performance and its robustness against overfitting can be attributed to both the fully convolutional network approach, which is already known to generalize well, and our multi-planar augmentation framework. The latter allows us to apply a single 2D model with fixed hyperparameters, resulting in a fully autonomous segmentation system of low computational complexity. Multi-planar training improves the generalization performance in several ways: 1) Sampling from multiple planes allows for a huge number of anatomically relevant images augmenting the training data; 2) Exposing a 2D model to multiple planes takes the 3D nature of the input into account while maintaining the statistical and computational efficiency of 2D kernels; 3) The systematic augmentation scheme allows test time augmentation to be performed, which increases the performance through variance reduction if errors across views are uncorrelated for a given subject (visualized in supplementary Fig. S.2). This makes the MPUnet an open source alternative to 3D fully convolutional neural networks.

11. **Bridging 2D and 3D Segmentation Networks for Computation-Efficient Volumetric Medical Image Segmentation: An Empirical Study of 2.5D Solutions**

- Authors: Yichi Zhang, Qingcheng Liao, Le Ding, Jicong Zhang
- Year: 2020 · Source: arXiv
- URL: https://arxiv.org/abs/2010.06163
- Relevance: **Direct empirical justification** — benchmarks majority voting, weighted voting and learned Volumetric Fusion-Net across axial/sagittal/coronal U-Nets; arXiv supports the project's majority-voting choice.

Abstract

Recently, deep convolutional neural networks have achieved great success for medical image segmentation. However, unlike segmentation of natural images, most medical images such as MRI and CT are volumetric data. In order to make full use of volumetric information, 3D CNNs are widely used. However, 3D CNNs suffer from higher inference time and computation cost, which hinders their further clinical applications. Additionally, with the increased number of parameters, the risk of overfitting is higher, especially for medical images where data and annotations are expensive to acquire. To issue this problem, many 2.5D segmentation methods have been proposed to make use of volumetric spatial information with less computation cost. Despite these works lead to improvements on a variety of segmentation tasks, to the best of our knowledge, there has not previously been a large-scale empirical comparison of these methods. In this paper, we aim to present a review of the latest developments of 2.5D methods for volumetric medical image segmentation. Additionally, to compare the performance and effectiveness of these methods, we provide an empirical study of these methods on three representative segmentation tasks involving different modalities and targets. Our experimental results highlight that 3D CNNs may not always be the best choice. Despite all these 2.5D methods can bring performance gains to 2D baseline, not all the methods hold the benefits on different datasets. We hope the results and conclusions of our study will prove useful for the community on exploring and developing efficient volumetric medical image segmentation methods.



Dataset	Metrics		2D CNNs			2.5D Multi-view Fusion		3D CNNs
		U-Net (A)	U-Net (S)	U-Net (C)	MV	WV	VFN	3D U-Net
Cardiac MRI	*Dice* ↑	0.8362	**0.8630**	0.8225	0.8992	0.8996	**0.9140**	0.9203
	95*HD* ↓	7.120	**3.274**	5.268	1.287	1.104	**1.021**	1.014
Spleen CT	*Dice* ↑	**0.9114**	0.8027	0.8858	0.9204	0.9208	**0.9410**	0.9452
	95*HD* ↓	**1.521**	8.710	2.165	1.511	1.508	**1.119**	1.222
Prostate MRI	*Dice* ↑	**0.7222**	0.4322	0.4545	0.6448	0.7490	**0.7731**	0.7677
	95*HD* ↓	**8.933**	23.72	15.31	4.956	4.402	**4.460**	5.935

Conclusion

In this paper, we aim at exploring 2.5D method for efficient volumetric medical image segmentation to overcome the lack of volumetric information of 2D CNNs and heavy computation cost of 3D CNNs, and make a large-scale empirical comparison on three representative medical image segmentation tasks involving different modalities and targets. Generally, 3D CNNs seem to be the best choice for segmentation accuracy, however, the unaffordable computational cost remains a hindrance for their further application. Besides, for anisotropic volumes that x and y axis preserve much higher resolution than the z axis, the performance advantages of 3D CNNs are not obvious, sometimes even worse than some 2.5D methods. For these anisotropic images with high discontinuity between slices, directly applying 3D CNNs may involve irrelevant features and hinder the learning process, which is in line with the conclusions in [15, 24]. For 2.5D segmentation methods, we observe that all these 2.5D methods have the potential to improve the performance of 2D baseline. Multi-view fusion of 2D results from different views can get segmentation results comparable to 3D CNNs. However, we still need to train 3 (or 4 using VFN) networks to get the final segmentation results. In addition, the improvement of this strategy for anisotropic volumes is not significant. For incorporating inter-slice information, we could obtain performance improvement while training only one network with slight computational increasement. Compared with naive multi-slice input to incorporate volumetric information, all RNN-based and attention-based 2.5D methods can further improve the segmentation performance. However, the performance gains of these methods are not consistent in different datasets. For fusing 2D/3D features, all these fusion methods can improve the segmentation efficiency compared with using only 2D CNNs, demonstrating the effectiveness of fusing 3D features to utilize volumetric information. However, the performance of different fusion methods is affected by the characteristics of different segmentation tasks. Table 5 presents the comparison of model parameters and average training time on different datasets (ratio to 2D baseline) of different segmentation methods. It can be observed that all 2.5D methods have fewer parameters and less computation cost compared with 3D CNNs. We cannot claim that we have completely reproduced all these methods, however, we tried our best to tune each method so as to ensure a fair comparison. Another limitation is that we only use 2D/3D U-Net as the backbone structure. However, they may not always be the best choice for all segmentation tasks. Still, we hope the results and conclusions of our study will prove useful for the community on exploring and developing efficient volumetric medical image segmentation methods.

12. **3D Semi-Supervised Learning with Uncertainty-Aware Multi-View Co-Training (UMCT)**

- Authors: Yingda Xia, Fengze Liu, Dong Yang, Jinzheng Cai, Lequan Yu, Zhuotun Zhu, Daguang Xu, Alan Yuille, Holger Roth
- Year: 2018 · Source: arXiv
- URL: https://arxiv.org/abs/1811.12506
- Relevance: Exploits view-consistency with uncertainty-weighted label fusion arXiv — motivates confidence-aware alternatives to plain voxel-level majority voting.

Abstract

While making a tremendous impact in various fields, deep neural networks usually require large amounts of labeled data for training which are expensive to collect in many applications, especially in the medical domain. Unlabeled data, on the other hand, is much more abundant. Semi-supervised learning techniques, such as co-training, could provide a powerful tool to leverage unlabeled data. In this paper, we propose a novel framework, uncertaintyaware multi-view co-training (UMCT), to address semisupervised learning on 3D data, such as volumetric data from medical imaging. In our work, co-training is achieved by exploiting multi-viewpoint consistency of 3D data. We generate different views by rotating or permuting the 3D data and utilize asymmetrical 3D kernels to encourage diversified features in different sub-networks. In addition, we propose an uncertainty-weighted label fusion mechanism to estimate the reliability of each view's prediction with Bayesian deep learning. As one view requires the supervision from other views in co-training, our self-adaptive approach computes a confidence score for the prediction of each unlabeled sample in order to assign a reliable pseudo label. Thus, our approach can take advantage of unlabeled data during training. We show the effectiveness of our proposed semi-supervised method on several public datasets from medical image segmentation tasks (NIH pancreas & LiTS liver tumor dataset). Meanwhile, a fully-supervised method based on our approach achieved state-of-the-art performances on both the LiTS liver tumor segmentation and the Medical Segmentation Decathlon (MSD) challenge, demonstrating the robustness and value of our framework, even when fully supervised training is feasible.

Conclusion

In this paper, we presented uncertainty-aware multi-view co-training (UMCT), aimed at 3D semi-supervised learning. We extended dual view co-training and deep cotraining on 2D images into multi-view 3D training, naturally introducing data-level view differences. We also proposed asymmetrical 3D kernels initialized from 2D pretrained models to introduce feature-level view differences. In multi-view settings, an uncertainty-weighted label fusion module (ULF) is built to estimate the accuracy of each view prediction by Bayesian uncertainty measurement. Epistemic uncertainty was estimated after transforming our model into a Bayesian deep network by adding dropout. This module gives a larger weight to more confident predictions and further boost the performance on multi-view predictions. Experiments under semi-supervised setting were performed on the NIH pancreas dataset and the LiTS liver tumor dataset. Other approaches were outperformed by a large margin on the NIH dataset. We also applied cotraining objectives on labeled data under fully supervised settings. The results were also promising, illustrating the effectiveness of multi-view co-training on 2D-initialized networks.

13. Reinventing 2D Convolutions for 3D Medical Images (ACS Convolutions)

- Authors: Jiancheng Yang, Xiaoyang Huang, Yi He, Jingwei Xu, Canqian Yang, Guozheng Xu, Bingbing Ni
- Year: 2021 · Source: arXiv
- URL: https://arxiv.org/abs/1911.10477
- Relevance: Splits 2D kernels across the three anatomical views inside a single network arXiv — an elegant counterpoint to training three independent 3D models and voting.

Abstract

There have been considerable debates over 2D and 3D representation learning on 3D medical images. 2D approaches could benefit from large-scale 2D pretraining, whereas they are generally weak in capturing large 3D contexts. 3D approaches are natively strong in 3D contexts, however few publicly available 3D medical dataset is large and diverse enough for universal 3D pretraining. Even for hybrid (2D + 3D) approaches, the intrinsic disadvantages within the 2D / 3D parts still exist. In this study, we bridge the gap between 2D and 3D convolutions by reinventing the 2D convolutions. We propose ACS (axial-coronal-sagittal) convolutions to perform natively 3D representation learning, while utilizing the pretrained weights on 2D datasets. In ACS convolutions, 2D convolution kernels are split by channel into three parts, and convoluted separately on the three views (axial, coronal and sagittal) of 3D representations. Theoretically, ANY 2D CNN (ResNet, DenseNet, or DeepLab) is able to be converted into a 3D ACS CNN, with pretrained weight of a same parameter size. Extensive experiments on several medical benchmarks (including classification, segmentation and detection tasks) validate the consistent superiority of the pretrained ACS CNNs, over the 2D / 3D CNN counterparts with / without pretraining. Even without pretraining, the ACS convolution can be used as a plugand-play replacement of standard 3D convolution, with smaller model size and less computation.

Table

Methods	Venue	Slices	Sens@0.5	Sens@1	Sens@2	Sens@4	Sens@8	Sens@16	Avg.[0.5,1,2,4]
3DCE [67]	MICCAI'18	×27	62.48	73.37	80.70	85.65	89.09	91.06	75.55
ULDor [68]	ISBI'19	×1	52.86	64.8	74.84	84.38	87.17	91.8	69.22
Volumetric Attention [69]	MICCAI'19	×3	69.10	77.90	83.80	-	-	-	-
Improved RetinaNet [70]	MICCAI'19	×3	72.15	80.07	86.40	90.77	94.09	96.32	82.35
MVP-Net [71]	MICCAI'19	×3	70.01	78.77	84.71	89.03	-	-	80.63
MVP-Net [71]	MICCAI'19	×9	73.83	81.82	87.60	91.30	-	-	83.64
MULAN (Mask R-CNN) [8]	MICCAI'19	×9	76.12	83.69	88.76	92.30	94.71	95.64	85.22
MULAN (Mask R-CNN) w/o SRL [8]	MICCAI'19	×9	-	-	-	-	-	-	84.22
2.5D Mask R-CNN **r.**	Ours	×3	70.34	79.11	86.16	90.94	94.04	96.01	81.64
2.5D Mask R-CNN **p.**	Ours	×3	72.57	79.89	86.80	91.04	94.24	96.32	82.58
3D Mask R-CNN **r.**	Ours	×3	63.58	74.63	82.88	88.03	91.39	93.99	77.28
3D Mask R-CNN **p.** I3D [56]	Ours	×3	72.01	80.09	86.54	91.29	93.91	95.68	82.48
ACS Mask R-CNN **r.**	Ours	×3	72.52	80.85	87.10	91.05	94.39	96.12	82.88
ACS Mask R-CNN **p.**	Ours	×3	73.00	81.17	87.05	91.78	94.63	95.48	83.25
2.5D Mask R-CNN **r.**	Ours	×7	73.37	81.13	86.73	90.96	93.99	95.79	83.05
2.5D Mask R-CNN **p.**	Ours	×7	73.66	82.15	87.72	91.38	93.86	95.98	83.73
3D Mask R-CNN **r.**	Ours	×7	64.10	75.14	82.13	87.44	91.53	94.22	77.20
3D Mask R-CNN **p.** I3D [56]	Ours	×7	75.37	83.43	88.68	92.20	94.52	96.07	84.92
ACS Mask R-CNN **r.**	Ours	×7	78.01	84.75	88.97	91.76	93.79	95.26	85.87
ACS Mask R-CNN **p.**	Ours	×7	**78.38**	**85.39**	**90.07**	**93.19**	**95.18**	**96.75**	**86.76**

Conclusion

We propose ACS convolution for 3D medical images, as a generic and plug-and-play replacement of standard 3D convolution. It enables pretraining from 2D images, which consistently provides singificant performance boost in our experiments. Even without pretraining, the ACS convolution is comparable or even better than 3D convolution, with smaller model size and less computation. In further study, we will focus on optimal ACS kernel axis assignment and integration with other 2D-to-3D transfer learning operators.

14. **Anisotropic 3D Multi-Stream CNN for Accurate Prostate Segmentation from Multi-Planar MRI**

- Authors: Anneke Meyer, Alireza Mehrtash, Marko Rak, Daniel Schindele, Martin Schostak, Clare Tempany, Tina Kapur, Peter Pieper, Christian Hansen
- Year: 2018 · Source: IEEE Xplore (ISBI 2018)
- URL: https://ieeexplore.ieee.org/document/8363549
- Relevance: Demonstrates empirically that adding sagittal and coronal streams improves anisotropic-volume segmentation ScienceDirect — quantitative evidence for multi-planar fusion.

Abstract

Individualized and accurate segmentations of the prostate are essential for diagnosis as well as therapy planning in prostate cancer (PCa). Most of the previously proposed prostate segmentation approaches rely purely on axial MRI scans, which suffer from low out-of-plane resolution. We propose a method that makes use of sagittal and coronal MRI scans to improve the accuracy of segmentation. These scans are typically acquired as standard of care for PCa staging, but are generally ignored by the segmentation algorithms. Our method is based on a multi-stream 3D convolutional neural network for the automatic extraction of isotropic high resolution segmentations from MR images. We evaluated segmentation performance on an isotropic high resolution ground truth (n = 40 subjects). The results show that the use of multi-planar volumes for prostate segmentation leads to improved segmentation results not only for the whole prostate (92.1% Dice similarity coefficient), but also in apex and base regions.

Conclusion

The aim of this work is to provide a simple yet powerful framework for the accurate segmentation and surface extraction of the prostate by use of axial, sagittal and coronal volumes. We proposed a 3D multi-stream CNN that simultaneously computes an isotropic high resolution segmentation from the three orthogonal volumes that can be easily used for surface extraction. We evaluated the segmentations against an isotropic manual ground truth that takes all three orthogonal volumes into account. Results show that an accurate segmentation can be obtained with this proposed method, improving the outcome compared to a single stream network using only axial volumes of the prostate. The created ground truth will be made available online for other researchers.

To the best of our knowledge, the only approach that utilizes the multi-planar volumes in prostate segmentation with deep learning is the one proposed by Cheng *et al*. [12]. Our method offers several advantages over this approach. First, by using a single network, as compared to the three networks, we can simplify the training process. Second, computationally expensive surface extraction methods, such as ball pivoting and

Poisson surface reconstruction, are not required: the surface can be easily generated from the high resolution segmentation. Consequently, our approach leads to faster prostate model generation while obtaining accurate results.

Because orthogonal volumes are usually included in the PCa MRI protocols, this work does not require any modification to the imaging procedures, and should encourage efforts to incorporate multi-planar volumes into the segmentation process for improved precision of prostate model generation for individualized therapy planning. Future work should include the enhancement of the networks architecture by use of residual connections. Additionally, automatic registration should be employed for cases where the volumes are not well aligned. Finally, a more clinical-relevant evaluation should be conducted, incorporating the effect of segmentation on the registration accuracy for targeted fusion biopsy procedures.

nnU-Net framework and extensions

15. nnU-Net: Self-adapting Framework for U-Net-Based Medical Image Segmentation

- Authors: Fabian Isensee, Jens Petersen, Andre Klein, David Zimmerer, Paul F. Jaeger, Simon Kohl, Jakob Wasserthal, Gregor Koehler, Tobias Norajitra, Sebastian Wirkert, Klaus H. Maier-Hein
- Year: 2018 · Source: arXiv
- URL: https://arxiv.org/abs/1809.10486
- Relevance: **Foundational** — the original nnU-Net framework (self-adapting 2D/3D U-Net arXiv with 5-fold cross-validation ensembling) used as the backbone in this project.

Abstract

The U-Net was presented in 2015. With its straight-forward and successful architecture it quickly evolved to a commonly used benchmark in medical image segmentation. The adaptation of the U-Net to novel problems, however, comprises several degrees of freedom regarding the exact architecture, pre-processing, training and inference. These choices are not independent of each other and substantially impact the overall performance. The present paper introduces the nnU-Net (”nonew-Net”), which refers to a robust and self-adapting framework on the basis of 2D and 3D vanilla U-Nets. We argue the strong case for taking away superfluous bells and whistles of many proposed network designs and instead focus on the remaining aspects that make out the performance and generalizability of a method. We evaluate the nnU-Net in the context of the Medical Segmentation Decathlon challenge, which measures segmentation performance in ten disciplines comprising distinct entities, image modalities, image geometries and dataset sizes, with no manual adjustments between datasets allowed. At the time of manuscript submission, nnU-Net achieves the highest mean dice scores across all classes and seven phase 1 tasks (except class 1 in BrainTumour) in the online leaderboard of the challenge.

Introduction

Medical Image Segmentation is currently dominated by deep convolutional neural networks (CNNs). However, each segmentation benchmark seems to require specialized architectures and training scheme modifications to achieve competitive performance [1,2,3,4,5]. This results in huge amounts of publications in the field that, alongside often limited validation on only few or even just a single dataset, make it increasingly difficult for researchers to identify methods that live up to their promised superiority beyond the limited scenarios they are demonstrated on. The Medical Segmentation Decathlon is intended to specifically address this issue: participants in this challenge are asked to create a segmentation algorithm that generalizes across 10 datasets corresponding to different entities of the human body. These algorithms may dynamically adapt to the specifics of a particular dataset, but are only allowed to do so in a fully automatic manner. The challenge is split into two successive phases: 1) a development phase in which participants are given access to 7 datasets to optimize their approach on and, using their final and thus frozen method, must submit segmentations for the corresponding 7 held-out test sets. 2) a second phase to evaluate the same exact method on 3 previously undisclosed datasets. We hypothesize that some of the architectural modifications presented recently are in part overfitted to specific problems or could suffer from imperfect validation that results from sub-optimal reimplementations of the state-of-theart. Using the U-Net as a benchmark on an in-house dataset, for example, requires the adaptation of the method to the novel problem. This spans several degrees of freedom. Even though the architecture itself is quite straight-forward, and even though the method is quite commonly used as a benchmark, we believe that the remaining interdependent choices regarding the exact architecture, preprocessing, training, inference and post-processing quite often cause the U-Net to underperform when used as a benchmark. Additionally, architectural tweaks that are intended to improve the performance of a network can rather easily be demonstrated to work if the network is not yet fully optimized for the task at hand, allowing for plenty of headroom for the tweak to improve results. In our own preliminary experiments, these tweaks however were unable to improve segmentation results in fully optimized networks and thus most likely unable to advance the state of the art. This leads us to believe that the influence of non-architectural aspects in segmentation methods is much more impactful, but at the same time also severely underestimated. In this paper, we present the nnU-Net (”no-new-Net”) framework. It resides on a set of three comparatively simple U-Net models that contain only minor modifications to the original U-Net [6]. We omit recently proposed extensions such as for example the use of residual connections [7,8], dense connections [5] or attention mechanisms [4]. The nnU-Net automatically adapts its architectures to the given image geometry. More importantly though, the nnU-Net framework thoroughly defines all the other steps around them. These are steps where much of the nets' performance can be gained or respectively lost: preprocessing (e.g. resampling and normalization), training (e.g. loss, optimizer setting and data augmentation), inference (e.g. patch-based strategy and ensembling across testtime augmentations and models) and a potential post-processing (e.g. enforcing single connected components if applicable).

Methods

Medical images commonly encompass a third dimension, which is why we consider a pool of basic U-Net architectures consisting of a 2D U-Net, a 3D U-Net and a U-Net Cascade. While the 2D and 3D U-Nets generate segmentations at full resolution, the cascade first generates low resolution segmentations and subsequently refines them.

Our architectural modifications as compared to the U-Net's original formulation are close to negligible and instead we focus our efforts on designing an automatic training pipeline for these models. The U-Net [6] is a successful encoder-decoder network that has received a lot of attention in the recent years. Its encoder part works similarly to a traditional classification CNN in that it successively aggregates semantic information at the expense of reduced spatial information. Since in segmentation, both semantic as well as spatial information are crucial for the success of a network, the missing spatial information must somehow be recovered. The U-Net does this through the decoder, which receives semantic information from the bottom of the 'U' and recombines it with higher resolution feature maps obtained directly from the encoder through skip connections. Unlike other segmentation networks, such as FCN [9] and previous iterations of DeepLab [10] this allows the U-Net to segment fine structures particularly well. Just like the original U-Net, we use two plain convolutional layers between poolings in the encoder and transposed convolution operations in the decoder. We deviate from the original architecture in that we replace ReLU activation functions with leaky ReLUs (neg. slope 1e −2) and use instance normalization [11] instead of the more popular batch normalization [12]. 2D U-Net Intuitively, using a 2D U-Net in the context of 3D medical image segmentation appears to be suboptimal because valuable information along the z-axis cannot be aggregated and taken into consideration. However, there is evidence [13] that conventional 3D segmentation methods deteriorate in performance if the dataset is anisotropic (cf. Prostate dataset of the Decathlon challenge). 3D U-Net A 3D U-Net seems like the appropriate method of choice for 3D image data. In an ideal world, we would train such an architecture on the entire patient's image. In reality however, we are limited by the amount of available GPU memory which allows us to train this architecture only on image patches. While this is not a problem for datasets comprised of smaller images (in terms of number of voxels per patient) such as the Brain Tumour, Hippocampus and Prostate datasets of this challenge, patch-based training, as dictated by datasets with large images such as Liver, may impede training. This is due to the limited field of view of the architecture which thus cannot collect sufficient contextual information to e.g. correctly distinguish parts of a liver from parts of other organs. U-Net Cascade To address this practical shortcoming of a 3D U-Net on datasets with large image sizes, we additionally propose a cascaded model. Therefore, a 3D U-Net is first trained on downsampled images (stage 1). The segmentation results of this U-Net are then upsampled to the original voxel spacing and passed as additional (one hot encoded) input channels to a second 3D U-Net, which is trained on patches at full resolution (stage 2). See Figure 1.

Dynamic adaptation of network topologies Due to the large differences in image size (median shape 482 × 512 × 512 for Liver vs. 36 × 50 × 35 for Hippocampus) the input patch size and number of pooling operations per axis (and thus implicitly the number of convolutional layers) must be automatically adapted for each dataset to allow for adequate aggregation of spatial information. Apart from adapting to the image geometries, there are technical constraints like the available memory to account for. Our guiding principle in this respect is to dynamically trade off the batch-size versus the network capacity, presented in detail below: We start out with network configurations that we know to be working with our hardware setup. For the 2D U-Net this configuration is an input patch size of 256×256, a batch size of 42 and 30 feature maps in the highest layers (number of feature maps doubles with each downsampling). We automatically adapt these parameters to the median plane size of each dataset (where we use the plane with the lowest in-plane spacing, corresponding to the highest resolution), so that the network effectively trains on entire slices. We configure the networks to pool along each axis until the feature map size for that axis is smaller than 8 (but not more than a maximum of 6 pooling operations). Just like the 2D U-Net, our 3D U-Net uses 30 feature maps at the highest resolution layers. Here we start with a base configuration of input patch size 128 × 128 × 128, and a batch size of 2. Due to memory constraints, we do not increase the input patch volume beyond 1283 voxels, but instead match the aspect ratio of the input patch size to that of the median size of the dataset in voxels. If the median shape of the dataset is smaller than 1283 then we use the median shape as input patch size and increase the batch size (so that the total number of voxels processed is the same as with 128 × 128 × 128 and a batch size of 2). Just like for the 2D U-Net we pool (for a maximum of 5 times) along each axis until the feature maps have size 8. For any network we limit the total number of voxels processed per optimizer step (defined as the input patch volume times the batch size) to a maximum of 5% of the dataset. For cases in excess, we reduce the batch size (with a lowerbound of 2).

Experiments and results

We optimize our network topologie using five-fold cross-validations on the phase 1 datasets. Our phase 1 cross-validation results as well as the corresponding submitted test set results are summarized in Table 2. - indicates that the U-Net Cascade was not applicable (i.e. necessary, according to our criteria) to a dataset because it was already fully covered by the input patch size of the 3D U-Net. The model that was used for the final submission is highlighted in bold. Although several test set submissions were allowed by the platform, we believe it to be bad practice to do so. Hence we only submitted once and report the results of this single submission. As can be seen in Table 2 our phase 1 cross-validation results are robustly recovered on the held-out test set indicating a desired absence of over-fitting. The only dataset that suffers from a dip in performance on all of its foreground classes is BrainTumour. The data of this phase 1 dataset stems from the BRATS challenge [16] for which such performance drops between validation and testing are a common sight and attributed to a large shift in the respective data and/or ground-truth distributions.

Table

label	1	2	3	1	1	2	1	2	1	2	1	1	2
2D U-Net	78.60	58.65	77.42	91.36	94.37	53.94	88.52	86.70	61.98	84.31	52.68	74.70	35.41
3D U-Net	**80.71**	**62.22**	**79.07**	92.45	94.11	61.74	**89.87**	**88.20**	60.77	83.73	55.87	77.69	42.69
3D U-Net stage1 only (U-Net Cascade)	-	-	-	90.63	94.69	47.01	-	-	-	-	65.33	79.45	49.65
3D U-Net (U-Net Cascade)	-	-	-	92.40	95.38	58.49	-	-	-	-	**66.85**	**79.30**	**52.12**
ensemble 2D U-Net+ 3D U-Net	80.79	61.72	79.16	**92.70**	94.30	60.24	89.78	88.09	**63.78**	**85.31**	55.96	78.26	40.46
ensemble 2D U-Net+ 3D U-Net (U-Net Cascade)	-	-	-	92.64	95.31	60.09	-	-	-	-	61.18	78.79	45.46
ensemble 3D U-Net+ 3D U-Net (U-Net Cascade)	-	-	-	92.63	**95.43**	**61.82**	-	-	-	-	65.16	79.70	49.14
test set	67.71	**47.73**	**68.16**	**92.77**	**95.24**	**73.71**	**90.37**	**88.95**	**75.81**	**89.59**	**69.20**	**79.53**	**52.27**

Table 2. Mean dice scores for the proposed models in all phase 1 tasks. All experiments were run as five-fold cross-validation. The models that we used for generating our test set submission are highlighted in bold. The dice scores of the test sets are shown at the bottom of the table. Test dice scores in bold denote that at the time of manuscript submission these scores were the highest in the online leaderboard of the challenge (decathlon.grand-challenge.org/evaluation/results).

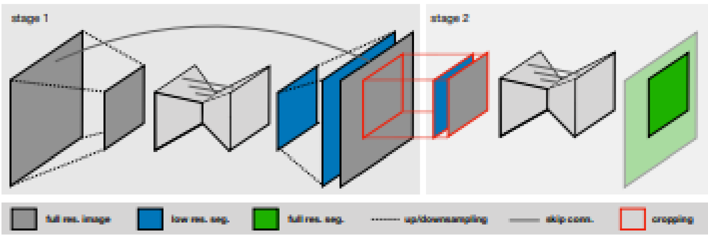


Fig. 1. U-Net Cascade (on applicable datasets only). Stage 1 (left): a 3D U-Net processes downsampled data, the resulting segmentation maps are upsampled to the original resolution. Stage 2 (right): these segmentations are concatenated as one-hot encodings to the full resolution data and refined by a second 3D U-Net.

Conclusion

In this paper we present the nnU-Net segmentation framework for the medical domain that directly builds around the original U-Net architecture [6] and dynamically adapts itself to the specifics of any given dataset. Based on our hypothesis that non-architectural modifications can be much more powerful than some of the recently presented architectural modifications, the essence of this framework is a thorough design of adaptive preprocessing, training scheme and inference. All design choices required to adapt to a new segmentation task are done in a fully automatic manner with no manual interaction. For each task the nnU-Net automatically runs a five-fold cross-validation for three different automatically configures U-Net models and the model (or ensemble) with the highest mean foreground dice score is chosen for final submission. In the context of the Medical Segmentation Decathlon we demonstrate that the nnU-Net performs competitively on the held-out test sets of 7 highly distinct medical datasets, achieving the highest mean dice scores for all classes of all tasks (except class 1 in the BrainTumour dataset) on the online leaderboard at the time of manuscript submission. We acknowledge that training three models and picking the best one for each dataset independently is not the cleanest solution. Given a larger time-scale, one could investigate proper heuristics to identify the best model for a given dataset prior to training. Our current tendency favors the U-Net Cascade (or the 3D U-Net if the cascade cannot be applied) with the sole (close) exceptions being the Prostate and Liver tasks. Additionally, the added benefit of many of our design choices, such as the use of Leaky ReLUs instead of regular ReLUs and the parameters of our data augmentation were not properly validated in the context of this challenge. Future work will therefore focus on systematically evaluating all design choices via ablation studies.

16. **nnU-Net: A Self-Configuring Method for Deep Learning-Based Biomedical Image Segmentation (expanded preprint of the Nature Methods 2021 paper)**

- Authors: Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, Klaus H. Maier-Hein
- Year: 2019/2020 · Source: arXiv
- URL: https://arxiv.org/abs/1904.08128
- Relevance: Full technical description of the automated fingerprint-based pipeline configuration arXiv used by nnU-Net v2 DNB (preprocessing, topology, training, post-processing).

Abstract

Biomedical imaging is a driver of scientific discovery and core component of medical care, currently stimulated by the field of deep learning. While semantic segmentation algorithms enable 3D image analysis and quantification in many applications, the design of respective specialised solutions is non-trivial and highly dependent on dataset properties and hardware conditions. We propose nnU-Net, a deep learning framework that condenses the current domain knowledge and autonomously takes the key decisions required to transfer a basic architecture to different datasets and segmentation tasks. Without manual tuning, nnU-Net surpasses most specialised deep learning pipelines in 19 public international competitions and sets a new state of the art in the majority of the 49 tasks. The results demonstrate a vast hidden potential in the systematic adaptation of deep learning methods to different datasets. We make nnU-Net publicly available as an open-source tool that can effectively be used out-of-the-box, rendering state of the art segmentation accessible to non-experts and catalyzing scientific progress as a framework for automated method design.

17. **nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation**

- Authors: Fabian Isensee, Tassilo Wald, Constantin Ulrich, Michael Baumgartner, Saikat Roy, Klaus Maier-Hein, Paul F. Jaeger
- Year: 2024 · Source: arXiv
- URL: https://arxiv.org/abs/2404.09556
- Relevance: Introduces **residual-encoder presets** for nnU-Net v2 and shows rigorous configuration/ensembling — not novel architectures — still drives SOTA; arXiv the core modern methodological reference.

Abstract

The release of nnU-Net marked a paradigm shift in 3D medical image segmentation, demonstrating that a properly configured U-Net architecture could still achieve state-of-the-art results. Despite this, the pursuit of novel architectures, and the respective claims of superior performance over the U-Net baseline, continued. In this study, we demonstrate that many of these recent claims fail to hold up when scrutinized for common validation shortcomings, such as the use of inadequate baselines, insufficient datasets, and neglected computational resources. By meticulously avoiding these pitfalls, we conduct a thorough and comprehensive benchmarking of current segmentation methods including CNNbased, Transformer-based, and Mamba-based approaches. In contrast to current beliefs, we find that the recipe for state-of-the-art performance is 1) employing CNN-based U-Net models, including ResNet and ConvNeXt variants, 2) using the nnU-Net framework, and 3) scaling models to modern hardware resources. These results indicate an ongoing innovation bias towards novel architectures in the field and underscore the need for more stringent validation standards in the quest for scientific progress.

Table

Table 1. Benchmark results of prevalent 3D medical segmentation methods measured as DSC score [%]. Colors are greener for higher scores, normalized per column. Abbreviations, "RT": Runtime measured in GPU hours on A100 40GB PCIe, "Arch.": Architecture, "TF": Transformer, "Mam": Mamba, "A3DS": Auto3DSeg, "AC": Auto-configuration, "nnU": implemented in nnU-Net framework, "DSC": Dice Similarity Coefficient.

	BTCV n=30	ACDC n=200	LiTS n=131	BraTS n=1251	KiTS n=489	AMOS n=360	VRAM [GB]	RT [h]	Arch.	nnU
nnU-Net (org.) [21]	83.08	91.54	80.09	91.24	86.04	88.64	7.70	9	CNN	Yes
nnU-Net ResEnc M	83.31	91.99	80.75	91.26	86.79	88.77	9.10	12	CNN	Yes
nnU-Net ResEnc L	83.35	91.69	81.60	91.13	88.17	89.41	22.70	35	CNN	Yes
nnU-Net ResEnc XL	83.28	91.48	81.19	91.18	**88.67**	89.68	36.60	66	CNN	Yes
MedNeXt L k3 [31]	84.70	**92.65**	82.14	91.35	88.25	89.62	17.30	68	CNN	Yes
MedNeXt L k5 [31]	**85.04**	92.62	**82.34**	**91.50**	87.74	**89.73**	18.00	233	CNN	Yes
STU-Net S [20]	82.92	91.04	78.50	90.55	84.93	88.08	5.20	10	CNN	Yes
STU-Net B [20]	83.05	91.30	79.19	90.85	86.32	88.46	8.80	15	CNN	Yes
STU-Net L [20]	83.36	91.31	80.31	91.26	85.84	89.34	26.50	51	CNN	Yes
SwinUNETR [32]	78.89	91.29	76.50	90.68	81.27	83.81	13.10	15	TF	Yes
SwinUNETRV2 [17]	80.85	92.01	77.85	90.74	84.14	86.24	13.40	15	TF	Yes
nnFormer [41]	80.86	92.40	77.40	90.22	75.85	81.55	5.70	8	TF	Yes
CoTr [37]	81.95	90.56	79.10	90.73	84.59	88.02	8.20	18	TF	Yes
No-Mamba Base	83.69	91.89	80.57	91.26	85.98	89.04	12.0	24	CNN	Yes
U-Mamba Bot [26]	83.51	91.79	80.40	91.26	86.22	89.13	12.40	24	Mam	Yes
U-Mamba Enc [26]	82.41	91.22	80.27	90.91	86.34	88.38	24.90	47	Mam	Yes
A3DS SegResNet [1,28]	80.69	90.69	79.28	90.79	81.11	87.27	20.00	22	CNN	No
A3DS DiNTS [1,18]	78.18	82.97	69.05	87.75	65.28	82.35	29.20	16	CNN	No
A3DS SwinUNETR [1,32]	76.54	82.68	68.59	89.90	52.82	85.05	34.50	9	TF	No

Conclusion

Our benchmark reveals a concerning trend in 3D medical image segmentation: most methods introduced in recent years fail to surpass the original nnU-Net baseline introduced in 2018. This raises the question: How can we steer the field towards genuine progress? In this study, we link the observed shortcomings to a widespread lack of rigor in method validation. To counteract this, we introduce: 1) A systematic collection of validation pitfalls along with recommendations for their avoidance, 2) The release of updated standardized baselines facilitating meaningful method validation, 3) A strategy for measuring the suitability of datasets for method benchmarking. Beyond these contributions, achieving true and lasting progress in the field requires a cultural shift, where the quality of validation is valued as much as the novelty of network architectures. Making this shift happen will be the responsibility of method developers, users, and reviewers alike.

18. **nnFormer: Interleaved Transformer for Volumetric Medical Image Segmentation**

- Authors: Hong-Yu Zhou, Jiansen Guo, Yinghao Zhang, Lequan Yu, Liansheng Wang, Yizhou Yu
- Year: 2021/2023 · Source: arXiv (also IEEE TIP 2023)
- URL: https://arxiv.org/abs/2109.03201
- Relevance: Transformer competitor benchmarked on MSD tasks — natural baseline for comparison against plane-wise nnU-Net v2.

Abstract

Transformer, the model of choice for natural language processing, has drawn scant attention from the medical imaging community. Given the ability to exploit long-term dependencies, transformers are promising to help atypical convolutional neural networks to overcome their inherent shortcomings of spatial inductive bias. However, most of recently proposed transformer-based segmentation approaches simply treated transformers as assisted modules to help encode global context into convolutional representations. To address this issue, we introduce nnFormer (i.e., not-another transFormer), a 3D transformer for volumetric medical image segmentation. nnFormer not only exploits the combination of interleaved convolution and self-attention operations, but also introduces local and global volume-based self-attention mechanism to learn volume representations. Moreover, nnFormer proposes to use skip attention to replace the traditional concatenation/summation operations in skip connections in U-Net like architecture. Experiments show that nnFormer significantly outperforms previous transformer-based counterparts by large margins on three public datasets. Compared to nnUNet, nnFormer produces significantly lower HD95 and comparable DSC results. Furthermore, we show that nnFormer and nnUNet are highly complementary to each other in model ensembling. Codes and models of nnFormer are available at https://git.io/JSf3i.

Figure

Methods	Average		WT		ET		TC		ED		NET	
	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑
nnUNet [40]	4.60	81.87	**3.64**	**91.99**	4.06	80.97	4.91	85.35	4.26	**84.39**	6.14	66.65
Our nnFormer	**4.42**	**82.02**	3.80	91.26	**3.87**	**81.80**	**4.49**	**86.02**	**4.17**	83.76	**5.76**	**67.29**
P-values	< 1e-2 (HD95), 8.8e-2 (DSC)											
nnAvg	4.09	82.65	3.43	92.33	3.69	82.26	4.17	86.14	3.92	84.95	5.23	67.55

(a) Brain tumor segmentation

Methods	Average		Aotra		Gallbladder		Kidney (Left)		Kidney (Right)		Liver		Pancreas		Spleen		Stomach	
	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑
nnUNet [40]	10.78	**86.99**	**5.91**	**93.01**	15.19	**71.77**	18.60	85.57	**6.44**	**88.18**	**1.62**	**97.23**	4.52	83.01	24.34	**91.86**	9.58	85.26
Our nnFormer	**10.63**	86.57	11.38	92.04	**11.55**	70.17	**18.09**	**86.57**	12.76	86.25	2.00	96.84	**3.72**	**83.35**	**16.92**	90.51	**8.58**	**86.83**
P-values	2e-2 (HD95), 7.7e-2 (DSC)																	
nnAvg	7.70	87.51	5.90	93.11	8.63	72.08	18.42	86.20	8.56	87.76	1.63	97.20	3.64	84.21	9.42	91.94	5.41	87.60

(b) Multi-organ segmentation (Synapse)

Methods	Average		RV		Myo		LV	
	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑	HD95 ↓	DSC ↑
nnUNet [40]	1.15	91.61	1.31	90.24	1.06	89.24	1.09	95.36
Our nnFormer	**1.12**	**92.06**	**1.23**	**90.94**	**1.04**	**89.58**	1.09	**95.65**
P-values	2e-2 (HD95), < 1e-2 (DSC)							
nnAvg	1.10	92.15	1.19	91.03	1.04	89.75	1.06	95.68

(c) Automated cardiac diagnosis (ACDC)

TABLE V: Comparison with nnUNet on three public datasets. **nnAvg** means that we simply average the predictions of nnUNet and nnFormer. Color green denotes the target result of nnAvg is the best among all three approaches. Besides, we also highlight the best results between nnUNet and nnFormer in bold font. We calculate p-values between the average performance of nnUNet and our nnFormer in both metrics on three public datasets.

#	Models	Average	RV	Myo	LV
0	1×LV-MSA + PM [3] + PE [3]	90.55	88.59	88.47	94.60
1	1×LV-MSA + PM [3] + Conv. Embed.	90.97	88.94	88.84	95.13
2	1×LV-MSA + Conv. Down. + Conv. Embed.	91.26	89.70	89.04	95.04
3	1×LV-MSA + 1×GV-MSA + Conv. Down. + Conv. Embed.	91.46	89.82	89.17	95.39
4	1×LV-MSA + 1×GV-MSA + Conv. Down. + Conv. Embed. + Skip Att.	91.85	90.41	89.50	95.63
5	1×LV-MSA + 1×SLV-MSA + 2×GV-MSA + Conv. Down. + Conv. Embed. + Skip Att.	92.06	90.94	89.58	95.65

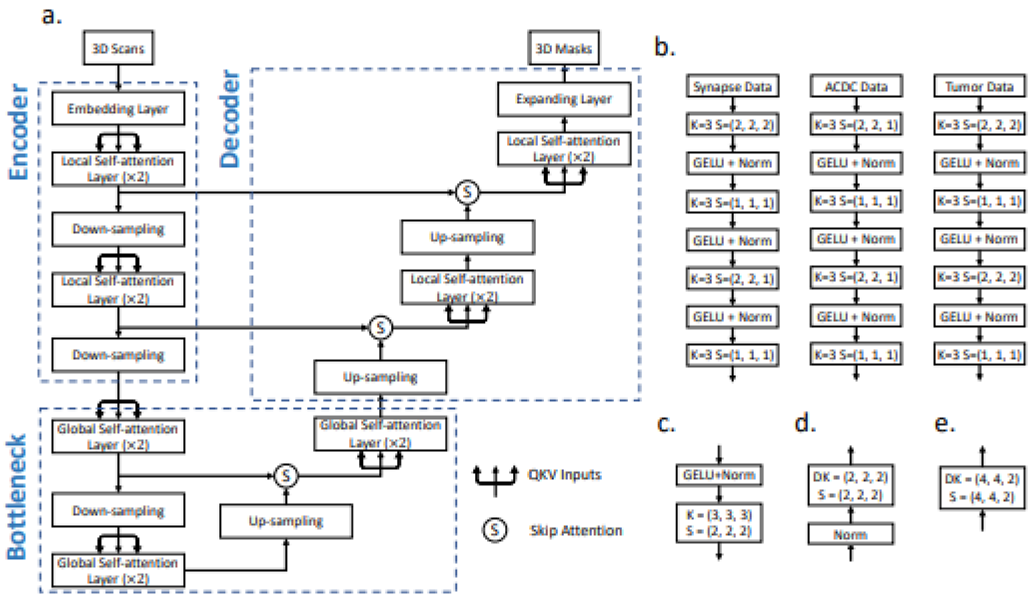


Fig. 2: Architecture of nnFormer. In (a), we show the overall architecture of nnFormer. In (b), we present more details of the embedding layers on three publicly available datasets. In (c), (d), (e), we display how to implement the down-sampling, up-sampling and expanding layers, respectively. In practice, the architecture may slightly vary depending on the input scan size. In (b)-(e), **K** denotes the convolutional kernel size, **DK** stands for the deconvolutional kernel size and **S** represents the stride. **Norm** refers to the layer normalization strategy.

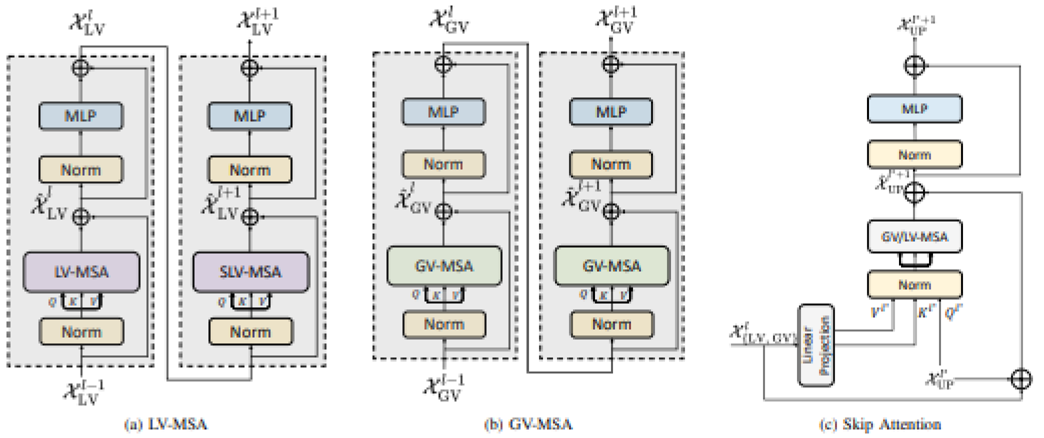


Fig. 3: Three types of attention mechanism in nnFormer. **Norm** denotes the layer normalization method. **MLP** is the abbreviation for multi-layer perceptron, which is a two-layer neural network in practice.

Conclusion

In this paper, we present a 3D transformer, nnFormer, for volumetric image segmentation. nnFormer is constructed on top of an interleaved stem of convolution and self-attention. Convolution helps encode precise spatial information and builds hierarchical object concepts. For self-attention, nnFormer employs three types of attention mechanism to entangle long-range dependencies. Specifically, local and global volume-based self-attention focus on constructing feature pyramids and providing large receptive field. Skip attention is responsible for bridging the gap between the encoder and decoder. Experiments show that nnFormer maintains great advantages over previous transformer-based models in both HD95 and DSC. Compared to nnUNet, nnFormer is significantly better in HD95 while producing comparable results in DSC.

More importantly, we demonstrate that nnFormer and nnUNet can be beneficial to each other in model ensembling, where the simple averaging operation can already produce great improvements.

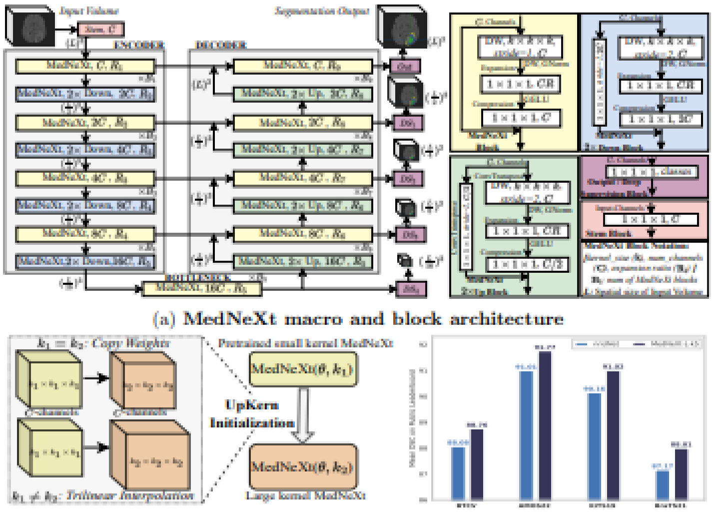
19. **MedNeXt: Transformer-driven Scaling of ConvNets for Medical Image Segmentation**

- Authors: Saikat Roy, Gregor Koehler, Constantin Ulrich, Michael Baumgartner, Jens Petersen, Fabian Isensee, Paul F. Jaeger, Klaus Maier-Hein
- Year: 2023 (MICCAI) · Source: arXiv
- URL: https://arxiv.org/abs/2303.09975
- Relevance: Fully ConvNeXt-style 3D encoder-decoder built *inside* the nnU-Net framework — an obvious architectural upgrade to test on Task08.

Abstract

There has been exploding interest in embracing Transformerbased architectures for medical image segmentation. However, the lack of large-scale annotated medical datasets make achieving performances equivalent to those in natural images challenging. Convolutional networks, in contrast, have higher inductive biases and consequently, are easily trainable to high performance. Recently, the ConvNeXt architecture attempted to modernize the standard ConvNet by mirroring Transformer blocks. In this work, we improve upon this to design a modernized and scalable convolutional architecture customized to challenges of data-scarce medical settings. We introduce MedNeXt, a Transformerinspired large kernel segmentation network which introduces – 1) A fully ConvNeXt 3D Encoder-Decoder Network for medical image segmentation, 2) Residual ConvNeXt up and downsampling blocks to preserve semantic richness across scales, 3) A novel technique to iteratively increase kernel sizes by upsampling small kernel networks, to prevent performance saturation on limited medical data, 4) Compound scaling at multiple levels (depth, width, kernel size) of MedNeXt. This leads to state-of-the-art performance on 4 tasks on CT and MRI modalities and varying dataset sizes, representing a modernized deep architecture for medical image segmentation. Our code is made publicly available at: https://github.com/MIC-DKFZ/MedNeXt.

Figure



(a) **MedNeXt macro and block architecture**

(b) **UpKern Initialization** (c) **Leaderboard Scores**

Fig. 1: **(a)** Architectural design of the MedNeXt. The network has 4 Encoder and Decoder layers each, with a bottleneck layer. MedNeXt blocks are present in Up and Downsampling layers as well. Deep Supervision is used at each decoder layer, with lower loss weights at lower resolutions. All residuals are *additive* while convolutions are padded to retain tensor sizes. **(b)** Upsampled Kernel (UpKern) initialization of a pair of MedNeXt architectures with similar configurations (θ) except kernel size (k_1, k_2). **(c)** MedNeXt-L ($5 \times 5 \times 5$) leaderboard performance.

Conclusion

In comparison to natural image analysis, medical image segmentation lacks architectures that benefit from scaling networks due to inherent domain challenges such as limited training data. In this work, MedNeXt is presented as a scalable Transformer-inspired fully-ConvNeXt 3D segmentation architecture customized for high performance on limited medical image datasets. We demonstrate MedNeXt's state-of-the-art performance across 4 challenging tasks against 7 strong baselines. Additionally, similar to ConvNeXt for natural images [22], we offer the compound scalable MedNeXt design as an effective modernization of standard convolution blocks for building deep networks for medical image segmentation.

clDice and topology-preserving loss functions

20. **clDice — A Novel Topology-Preserving Loss Function for Tubular Structure Segmentation**

- Authors: Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, Bjoern H. Menze
- Year: 2021 (CVPR) · Source: arXiv + IEEE Xplore
- URLs: https://arxiv.org/abs/2003.07311 · https://ieeexplore.ieee.org/document/9578225
- Relevance: **Core citation** — defines soft-clDice with homotopy-equivalence guarantees for tubular structures; Semantic ScholararXiv the exact loss used in this project.

Abstract

Accurate segmentation of tubular, network-like structures, such as vessels, neurons, or roads, is relevant to many fields of research. For such structures, the topology is their most important characteristic; particularly preserving connectedness: in the case of vascular networks, missing a connected vessel entirely alters the blood-flow dynamics. We introduce a novel similarity measure termed centerlineDice (short clDice), which is calculated on the intersection of the segmentation masks and their (morphological) skeleta. We theoretically prove that clDice guarantees topology preservation up to homotopy equivalence for binary 2D and 3D segmentation. Extending this, we propose a computationally efficient, differentiable loss function (soft-clDice) for training arbitrary neural segmentation networks. We benchmark the soft-clDice loss on five public datasets, including vessels, roads and neurons (2D and 3D). Training on soft-clDice leads to segmentation with more accurate connectivity information, higher graph similarity, and better volumetric scores.

Introduction

Segmentation of tubular and curvilinear structures is an essential problem in numerous domains, such as clinical and biological applications (blood vessel and neuron segmentation from microscopic, optoacoustic, or radiology images), remote sensing applications (road network segmentation from satellite images) and industrial quality control, etc. In the aforementioned domains, a topologically accurate segmentation is necessary to guarantee error-free downstream tasks, such as computational hemodynamics, route planning, Alzheimer's disease prediction [18], or stroke modeling [20]. When optimizing computational algorithms for segmenting curvilinear structures, the two most commonly used categories of quantitative performance measures for evaluating segmentation accuracy of tubular struc-tures, are 1) overlap based measures such as Dice, precision, recall, and Jaccard index; and 2) volumetric distance measures such as the Hausdorff and Mahalanobis distance [21, 42, 37, 16]. However, in most segmentation problems, where the object of interest is 1) locally a tubular structure and 2) globally forms a network, the most important characteristic is the connectivity of the global network topology. Note that network in this context implies a physically connected structure, such as a vessel network, a road network, etc., which is also the primary structure of interest for the given image data. As an example, one can refer to brain vasculature analysis, where a missed vessel segment in the segmentation mask can pathologically be interpreted as a stroke or may lead to dramatic changes in a global simulation of blood flow. On the other hand, limited over- or undersegmentation of vessel radius can be tolerated, because it does not affect clinical diagnosis. For evaluating segmentations in such tubular-network structures, traditional volume-based performance indices are sub-optimal. For example, Dice and Jaccard rely on the average voxel-wise hit or miss prediction [48]. In a task like network-topology extraction, a spatially contiguous sequence of correct voxel prediction is more meaningful than a spurious correct prediction. This ambiguity is relevant for objects of interest, which are of the same thickness as the resolution of the signal. For them, it is evident that a singlevoxel shift in the prediction can change the topology of the whole network. Further, a globally averaged metric does not equally weight tubular-structures with large, medium, and small radii (cf. Fig 1). In real vessel datasets, where vessels of wide radius ranges exist, e.g. 30 μm for arterioles and 5 μm for capillaries [50, 9], training on a globally averaged loss induces a strong bias towards the volumetric segmentation of large vessels. Both scenarios are pronounced in imaging modalities, such as fluorescence microscopy [50, 60] and optoacoustics, which focus on mapping small capillary structures. To this end, we are interested in a topology-aware image segmentation, eventually enabling a correct network extraction. Therefore, we ask the following research questions: Q1. What is a good pixelwise measure to benchmark segmentation algorithms for tubular, and related linear and curvilinear structure segmentation while guaranteeing the preservation of the network-topology? Q2. Can we use this improved measure as a loss function for neural networks, which is a void in existing literature?

Figures

Table 1. Quantitative experimental results for the Massachusetts road dataset (Roads), the CREMI dataset, the DRIVE retina dataset and the Vessap dataset (3D). Bold numbers indicate the best performance. The performance according to the *clDice* measure is highlighted in rose. For all experiments we observe that using *soft-clDice* in $\mathcal{L}_c$ results in improved scores compared to *soft-Dice*. This improvement holds for almost $\alpha > 0$; α can be interpreted as a dataset specific hyper-parameter.

Dataset	Network	Loss	Dice	Accuracy	*clDice*	β_0 Error	β_1 Error	SMD [4]	χ_{error}	Opt-J F1 [7]
Roads	FCN	*soft-dice*	64.84	95.16	70.79	1.474	1.408	0.1216	2.634	0.766
		$\mathcal{L}_c, \alpha = 0.1$	66.52	95.70	74.80	0.987	1.227	0.1002	2.625	0.768
		$\mathcal{L}_c, \alpha = 0.2$	**67.42**	**95.80**	76.25	**0.920**	1.280	**0.0954**	2.526	0.770
		$\mathcal{L}_c, \alpha = 0.3$	65.90	95.35	74.86	0.974	1.197	0.1003	2.448	0.775
		$\mathcal{L}_c, \alpha = 0.4$	67.18	95.46	**76.92**	0.934	**1.092**	0.0991	**2.183**	**0.803**
		$\mathcal{L}_c, \alpha = 0.5$	65.77	95.09	75.22	0.947	1.184	0.0991	2.361	0.782
	U-NET	*soft-dice*	76.23	96.75	86.83	0.491	1.256	0.0589	1.120	0.881
		$\mathcal{L}_c, \alpha = 0.1$	**76.66**	**96.77**	87.35	0.359	**0.938**	0.0457	0.980	0.878
		$\mathcal{L}_c, \alpha = 0.2$	76.25	96.76	87.29	**0.312**	1.031	**0.0415**	0.865	0.900
		$\mathcal{L}_c, \alpha = 0.3$	74.85	96.57	86.10	0.322	1.062	0.0504	0.827	0.913
		$\mathcal{L}_c, \alpha = 0.4$	75.38	96.60	86.16	0.344	1.016	0.0483	**0.755**	**0.916**
		$\mathcal{L}_c, \alpha = 0.5$	76.45	96.64	**88.17**	0.375	0.953	0.0527	1.080	0.894
	Mosinska et al.	[29, 17]	-	97.54	-	-	2.781	-	-	-
	Hu et al.	[17]	-	97.28	-	-	1.275	-	-	-
CREMI	U-NET	*soft-dice*	91.54	97.11	95.86	0.259	0.657	0.0461	1.087	0.904
		$\mathcal{L}_c, \alpha = 0.1$	91.76	**97.21**	96.05	0.222	0.556	**0.0395**	1.000	0.900
		$\mathcal{L}_c, \alpha = 0.2$	91.66	97.15	96.01	0.231	0.630	0.0419	0.991	0.902
		$\mathcal{L}_c, \alpha = 0.3$	**91.78**	97.18	**96.21**	**0.204**	**0.537**	0.0437	**0.919**	**0.913**
		$\mathcal{L}_c, \alpha = 0.4$	91.56	97.12	96.09	0.250	0.630	0.0444	0.995	0.902
		$\mathcal{L}_c, \alpha = 0.5$	91.66	97.16	96.16	0.231	0.620	0.0455	0.991	0.907
	Mosinska et al.	[29, 17]	82.30	94.67	-	-	1.973	-	-	-
	Hu et al.	[17]	-	94.56	-	-	1.113	-	-	-
DRIVE retina	FCN	*soft-Dice*	78.23	96.27	78.02	2.187	1.860	0.0429	3.275	0.773
		$\mathcal{L}_c, \alpha = 0.1$	78.36	96.25	79.02	2.100	1.610	0.0393	3.203	0.777
		$\mathcal{L}_c, \alpha = 0.2$	**78.75**	96.29	80.22	1.892	1.382	0.0383	2.895	0.793
		$\mathcal{L}_c, \alpha = 0.3$	78.29	96.20	80.28	1.888	**1.332**	**0.0318**	2.918	**0.798**
		$\mathcal{L}_c, \alpha = 0.4$	78.00	96.11	80.43	2.036	1.602	0.0423	3.141	0.764
		$\mathcal{L}_c, \alpha = 0.5$	77.76	96.04	**80.95**	**1.836**	1.408	0.0394	**2.848**	0.794
	U-Net	*soft-Dice*	74.25	95.63	75.71	1.745	1.455	0.0649	2.997	0.760
		$\mathcal{L}_c, \alpha = 0.5$	**75.21**	**95.82**	**76.86**	**1.538**	**1.389**	**0.0586**	**2.737**	**0.767**
	Mosinska et al.	[29, 17]	-	95.43	-	-	2.784	-	-	-
	Hu et al.	[17]	-	95.21	-	-	1.076	-	-	-
Vessap data	FCN, 1 ch	*soft-dice*	85.21	**96.03**	90.88	3.385	4.458	0.00459	5.850	0.862
		$\mathcal{L}_c, \alpha = 0.5$	**85.44**	95.91	**91.32**	**2.292**	**3.677**	**0.00417**	**5.620**	**0.864**
	FCN, 2 ch	*soft-dice*	85.31	95.82	90.10	2.833	4.771	0.00629	6.080	0.849
		$\mathcal{L}_c, \alpha = 0.1$	85.96	95.99	91.02	2.896	**4.156**	0.00447	5.980	0.860
		$\mathcal{L}_c, \alpha = 0.2$	**86.45**	**96.11**	91.22	2.656	4.385	0.00466	5.530	0.869
		$\mathcal{L}_c, \alpha = 0.3$	85.72	95.93	91.20	2.719	4.469	**0.00423**	5.470	0.866
		$\mathcal{L}_c, \alpha = 0.4$	85.65	95.95	**91.65**	2.719	4.469	**0.00423**	5.670	0.869
		$\mathcal{L}_c, \alpha = 0.5$	85.28	95.76	91.22	**2.615**	4.615	0.00433	**5.320**	**0.870**
	U-Net, 1 ch	*soft-dice*	87.46	96.35	91.18	3.094	5.042	0.00549	5.300	0.863
		$\mathcal{L}_c, \alpha = 0.5$	**87.82**	**96.52**	**93.03**	**2.656**	**4.615**	**0.00533**	**4.910**	**0.872**
	U-Net, 2 ch	*soft-dice*	87.98	96.56	90.16	2.344	4.323	0.00507	5.550	0.855
		$\mathcal{L}_c, \alpha = 0.1$	88.13	96.59	91.12	2.302	4.490	0.00465	5.180	**0.872**
		$\mathcal{L}_c, \alpha = 0.2$	87.96	96.74	92.52	2.208	**3.979**	0.00342	**4.830**	0.861
		$\mathcal{L}_c, \alpha = 0.3$	87.70	96.71	92.56	**2.115**	4.521	**0.00309**	5.260	0.858
		$\mathcal{L}_c, \alpha = 0.4$	**88.57**	**96.87**	**93.25**	2.281	4.302	0.00327	5.370	0.868
		$\mathcal{L}_c, \alpha = 0.5$	88.14	96.74	92.75	2.135	4.125	0.00328	5.390	0.864

Contributions

The objective of this paper is to identify an efficient, general, and intuitive loss function that enables topology preservation while segmenting tubular objects. We introduce a novel connectivity-aware similarity measure named clDice for benchmarking tubular-segmentation algorithms. Importantly, we provide theoretical guarantees for the topological correctness of the clDice for binary 2D and 3D segmentation. As a consequence of its formulation based on morphological skeletons, our measure pronounces the network's topology instead of equally weighting every voxel. Using a differentiable soft-skeletonization, we show that the clDice measure can be used to train neural networks. We show experimental results for various 2D and 3D network segmentation tasks to demonstrate the practical applicability of our proposed similarity measure and loss function.

Results & discussion

We trained two segmentation architectures, a U-Net and an FCN, for the various loss functions in our experimental setup. As a baseline, we trained the networks using soft-dice and compared it with the ones trained using the proposed loss (Eq. 3), by varying α from (0.1 to 0.5). Quantitative: We observe that including soft-clDice in any proportion ($\alpha > 0$) leads to improved topological, volumetric and graph similarity for all 2D and 3D datasets, see Table 1. We conclude that α can be interpreted as a hyper parameter which can be tuned per-dataset. Intuitively, increasing the α improves the clDice measure for most experiments. Most often, clDice is high or highest when the graph and topology based measures are high or highest, particularly the β_1 Error, Streetmover distance and Opt-J F1 score; quantitatively indicating that topological properties are indeed represented in the clDice measure. In spite of not optimizing for a high soft-clDice on the background class, all of our networks converge to superior segmentation results. This not only reinforces our assumptions on dataset-specific necessary conditions but also validates the practical applicability of our loss. Our findings hold for the different network architectures, for 2D or 3D, and for tubular or curvilinear structures, strongly indicating its generalizability to analogous binary segmentation tasks. Observe that CREMI and the synthetic vessel dataset (see Supplementary material) appear to have the smallest increase in scores over the baseline. We attribute this to them being the least complex datasets in the collection, with CREMI having an almost uniform thickness of radii and the synthetic data having a high signal-to-noise ratio and insignificant illumination variation. More importantly, we observe larger improvements for all measures in case of the more complex Vessap and Roads data see Figure 5. In direct comparison to performance measures reported in two recent publications by Hu et al. and Mosinska et al. [17, 29], we find that our approach is on par or better in terms of Accuracy and Betti Error for the Roads and CREMI dataset. It is important to note that we used a smaller subset of training data for the Road dataset compared to both while using the same test set. Hu et al. reported a Betti error for the DRIVE data, which exceeds ours; however, it is important to consider that their approach explicitly minimizes the mismatch of the persistence diagram, which has significantly higher computational complexity during training, see the section below. We find that our proposed loss performs superior to the baseline in almost every scenario. The improvement appears to be pronounced when evaluating the highly relevant graph and topology based measures, including the recently introduced OPT-Junction F1 by Citraro et al. [7]. Our results are consistent across different network architectures, indicating that soft-clDice can be deployed to any network architecture. Qualitative: In Figure 5, typical results for our datasets are depicted. Our networks trained on the proposed loss term recover connections, which were false negatives when trained with the soft-Dice loss. These missed connections appear to be particularly frequent in the complex road and DRIVE dataset. For the CREMI dataset, we observe these situations less frequently, which is in line with the very high quantitative scores on the CREMI data. Interestingly, in the real 3D vessel dataset, the soft-Dice loss oversegments vessels, leading to false positive connections. This is not the case when using our proposed loss function, which we attribute to its topology-preserving nature. Additional qualitative results can be inspected in the supplementary. Computational Efficiency: Naturally, inference times of CNNs with the same architecture but different training losses are identical. However, during training, our soft-skeleton algorithm requires O(kn2) complexity for an n × n 2D image where k is the number of iterations. As a comparison, [17] needs O(c 2mlog(m)) (see [15]) complexity to compute the 1d persistent homology where d is the number of points with zero gradients in the prediction and m is the number of simplices. Roughly, c is proportional to n 2 , and m is of O(n 2) for a 2D Euclidean grid. Thus, the worst complexity of [17] is O(n 6 log(n)). Additionally, their approach requires an O(clog(c)) complexity to find an optimal matching of the birth-death pairs. We note that the total run-time overhead for soft-clDice compared to soft-Dice is marginal, i.e., for batch-size of 4 and 1024x1024 image resolution, the former takes 1.35s while the latter takes 1.24s on average

Conclusion

We introduce clDice, a novel topology-preserving similarity measure for tubular structure segmentation. Importantly, we present a theoretical guarantee that clDice enforces topology preservation up to homotopy equivalence. Next, we use a differentiable version of the clDice, soft-clDice, in a loss function, to train state-of-the-art 2D and 3D neural networks. We use clDice to benchmark segmentation quality from a topology-preserving perspective along with multiple volumetric, topological, and graph-based measures. We find that training on soft-clDice leads to segmentations with more accurate connectivity information, better graph-similarity, better Euler characteristics, and improved Dice and Accuracy. Our soft-clDice is computationally efficient and can be readily deployed to any other deep learning-based segmentation tasks such as neuron segmentation in biomedical imaging, crack detection in industrial quality control, or remote sensing.

21. Topology-Preserving Deep Image Segmentation

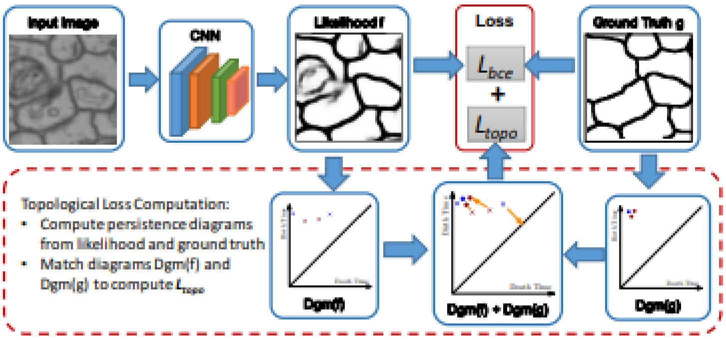
- Authors: Xiaoling Hu, Fuxin Li, Dimitris Samaras, Chao Chen
- Year: 2019 (NeurIPS) · Source: arXiv
- URL: https://arxiv.org/abs/1906.05404
- Relevance: First differentiable persistent-homology loss enforcing correct Betti numbers arXiv — the conceptual complement to clDice and a required comparison in topology-aware sections.

Abstract

Segmentation algorithms are prone to make topological errors on fine-scale structures, e.g., broken connections. We propose a novel method that learns to segment with correct topology. In particular, we design a continuous-valued loss function that enforces a segmentation to have the same topology as the ground truth, i.e., having the same Betti number. The proposed topology-preserving loss function is differentiable and we incorporate it into end-to-end training of a deep neural network. Our method achieves much better performance on the Betti number error, which directly accounts for the topological correctness. It also performs superiorly on other topology-relevant metrics, e.g., the Adjusted Rand Index and the Variation of Information. We illustrate the effectiveness of the proposed method on a broad spectrum of natural and biomedical datasets.

Introduction

Image segmentation, i.e., assigning labels to all pixels of an input image, is crucial in many computer vision tasks. State-of-the-art deep segmentation methods [23, 19, 8–10] learn high quality feature representations through an end-to-end trained deep network and achieve satisfactory per-pixel accuracy. However, these segmentation algorithms are still prone to make errors on fine-scale structures, such as small object instances, instances with multiple connected components, and thin connections. These fine-scale structures may be crucial in analyzing the functionality of the objects. For example, accurate extraction of thin parts such as ropes and handles is crucial in planning robot actions, e.g., dragging or grasping. In biomedical images, correct delineation of thin objects such as neuron membranes and vessels is crucial in providing accurate morphological and structural quantification of the underlying system. A broken connection or a missing component may only induce marginal per-pixel error, but can cause catastrophic functional mistakes. See Fig. 1 for an example. We propose TopoNet, a novel deep segmentation method that learns to segment with correct topology. In particular, we propose a topological loss that enforces the segmentation results to have the same topology as the ground truth, i.e., having the same Betti number (number of connected components and handles). A neural network trained with such loss will achieve high topological fidelity without sacrificing per-pixel accuracy. The main challenge in designing such loss is that topological information, namely, Betti numbers, are discrete values. We need a continuous-valued measurement of the topological similarity between a prediction and the ground truth; and such measurement needs to be differentiable in order to backpropagate through the network. To this end, we propose to use theory from computational topology [13], which summarizes the topological information from a continuous-valued function (in our case, the likelihood function f is predicted by a neural network). Instead of acquiring the segmentation by thresholding f at 0.5 and inspecting its topology, persistent homology [13, 14, 40] captures topological information carried by f over all possible thresholds. This provides a unified, differentiable approach of measuring the topological similarity between f and the ground truth, called the topological loss. We derive the gradient of the loss so that the network predicting the likelihood function can be optimized w.r.t. the topological loss. TopoNet is the first end-to-end deep segmentation network with guaranteed topological correctness. We show that when the topological loss is decreased to zero, the segmentation is guaranteed to be topologically correct, i.e., have identical topology as the ground truth. Our method is empirically validated by comparing with state-of-the-arts on natural and biomedical datasets with fine-scale structures. It achieves superior performance on metrics that encourage structural accuracy. In particular, our method significantly outperforms others on the Betti number error which exactly measures the topological accuracy. Fig. 1 shows a qualitative result. Our method demonstrates how topological computation and deep learning can be mutually beneficial. While our method empowers deep neural network with topological constraints, it also can be seen as a powerful approach on topological analysis in that the observation function is now learned with a highly nonlinear deep network. This enables topology to be estimated based on a semantically informed and denoised observation function. Related work. The closest method to ours is by Mosinska et al. [25], which also proposes a topologyaware loss. Instead of actually computing and comparing the topology, their approach uses the response of selected filters from a pretrained VGG19 network to construct the loss. These filters prefer elongated shapes and thus alleviate the broken connection issue. But this method is hard to generalize to more complex settings with connections of arbitrary shapes. Furthermore, even if this method achieves zero loss, its segmentation is not guaranteed to be topologically correct. Different ideas have been proposed to capture fine details of objects, mostly revolving around deconvolution and upsampling [23, 8–10, 27, 32]. However these methods focus on the prediction accuracy of individual pixels and are intrinsically topology-agnostic. Topological constraints, e.g., connectivity and loop-freeness, have been incorporated into variational [18, 22, 36, 33] and MRF/CRFbased segmentation methods [38, 28, 39, 6, 2, 35, 29, 15]. However, these methods focus on enforcing topological constraints in the inference stage, while the trained model is agnostic of the topological prior. In neuron image segmentation, some methods [17, 37] directly find an optimal partition of the image into neurons, and thus avoid segmenting membranes. These methods cannot be generalized to other problems when the object of interest may not be closed loops, e.g., vessels, cracks and roads. For completeness, we also refer to other existing works on extracting topological features and training kernel classifiers [1, 31, 21, 5, 7]. In graphics, topological similarity was used to simplify and align shapes [30]. As for deep neural networks, Hofer et al. [20] proposed a CNN-based topological classifier. This method directly extracts topological information from an input image/shape/graph as input for CNN, hence cannot generate segmentations that preserve topological priors learned from the training set. To the best of our knowledge, no existing work uses topological information as a loss for training an end-to-end deep neural network.



Conclusion

We introduce a new topological loss driven by persistent homology and incorporate it into end-to-end training of deep neural networks. Our method are particularly suitable for fine structure image segmentation. Quantitative and qualitative results show that the proposed topological loss term helps to achieve better performance in terms of topological-relevant metrics. Our proposed topological loss term is generic and can be incorporated into different architectures.

22. **A Topological Loss Function for Deep-Learning Based Image Segmentation Using Persistent Homology**

- Authors: James R. Clough, Nicholas Byrne, Ilkay Oksuz, Veronika A. Zimmer, Julia A. Schnabel, Andrew P. King
- Year: 2020 (IEEE TPAMI) · Source: arXiv
- URL: https://arxiv.org/abs/1910.01877
- Relevance: Prior-driven persistent-homology loss that does not require topology labels — alternative topology regulariser useful to benchmark against clDice.

Abstract

We introduce a method for training neural networks to perform image or volume segmentation in which prior knowledge about the topology of the segmented object can be explicitly provided and then incorporated into the training process. By using the differentiable properties of persistent homology, a concept used in topological data analysis, we can specify the desired topology of segmented objects in terms of their Betti numbers and then drive the proposed segmentations to contain the specified topological features. Importantly this process does not require any ground-truth labels, just prior knowledge of the topology of the structure being segmented. We demonstrate our approach in four experiments: one on MNIST image denoising and digit recognition, one on left ventricular myocardium segmentation from magnetic resonance imaging data from the UK Biobank, one on the ACDC public challenge dataset and one on placenta segmentation from 3-D ultrasound. We find that embedding explicit prior knowledge in neural network segmentation tasks is most beneficial when the segmentation task is especially challenging and that it can be used in either a semi-supervised or post-processing context to extract a useful training gradient from images without pixelwise labels.

Conclusion

We have presented a loss function for training CNNs to perform image segmentation which assesses the extent to which the proposed segmentation adheres to our prior knowledge of its topology. Using persistent homology, the robustness of various topological features can be computed in a manner which allows for gradient descent on the weights of the network performing the segmentation. Our experiments have shown that our approach is applicable to 2D images and 3D volumes, to CMR imaging and ultrasound, and that in these cases it improves the pixelwise and topological accuracy of the resulting segmentations.

23. **Skeleton Recall Loss for Connectivity Conserving and Resource Efficient Segmentation of Thin Tubular Structures**

- Authors: Yannick Kirchhoff, Maximilian R. Rokuss, Saikat Roy, Balint Kovacs, Constantin Ulrich, Tassilo Wald, Maximilian Zenk, Philipp Vollmuth, Jens Kleesiek, Fabian Isensee, Klaus Maier-Hein
- Year: 2024 (ECCV) · Source: arXiv
- URL: https://arxiv.org/abs/2404.03010
- Relevance: CPU-side skeleton recall loss that overcomes clDice's GPU memory cost on large 3D volumes — highly relevant because the project runs clDice on full-resolution CT.

Abstract

Accurately segmenting thin tubular structures, such as vessels, nerves, roads or concrete cracks, is a crucial task in computer vision. Standard deep learning-based segmentation loss functions, such as Dice or Cross-Entropy, focus on volumetric overlap, often at the expense of preserving structural connectivity or topology. This can lead to segmentation errors that adversely affect downstream tasks, including flow calculation, navigation, and structural inspection. Although current topology-focused losses mark an improvement, they introduce significant computational and memory overheads. This is particularly relevant for 3D data, rendering these losses infeasible for larger volumes as well as increasingly important multi-class segmentation problems. To mitigate this, we propose a novel Skeleton Recall Loss, which effectively addresses these challenges by circumventing intensive GPU-based calculations with inexpensive CPU operations. It demonstrates overall superior performance to current state-of-the-art approaches on five public datasets for topology-preserving segmentation, while substantially reducing computational overheads by more than 90%. In doing so, we introduce the first multi-class capable loss function for thin structure segmentation, excelling in both efficiency and efficacy for topology-preservation. Our code is available to the community, providing a foundation for further advancements, at: https://github.com/MIC-DKFZ/Skeleton-Recall.

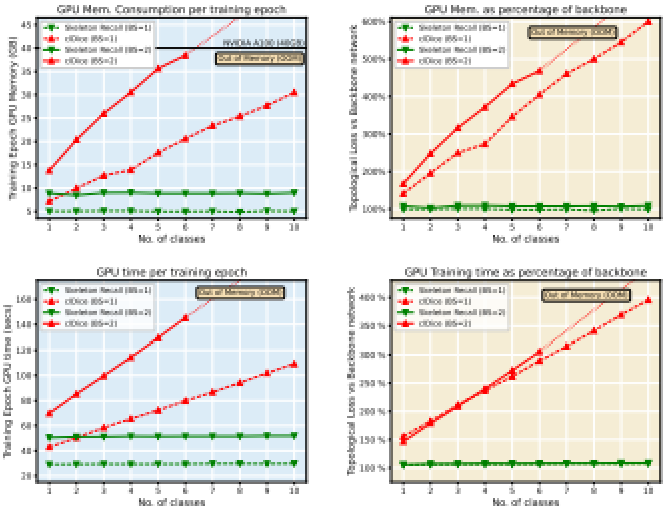


Fig. 7: Resource Utilization for Multi-class Segmentation. *Skeleton Recall Loss* requires minimal additional GPU memory and training time for an increasing number of classes, as it avoids a differentiable skeleton computation. Competing methods like *clDice Loss*, on the other hand, incur enormous overheads which make multi-class training on datasets such as TopCoW [40] infeasible. Analysis was performed for two different batch sizes (BS) on a single A100 40GB GPU, averaged over 5 training epochs, for ease of comparability.

Conclusion

This paper proposes a novel loss function, Skeleton Recall Loss, designed for connectivity preserving semantic segmentation. It is domain and architecture agnostic and, unlike existing methods, requires minimal additional training time and memory. Through extensive evaluation on five publicly available datasets, we demonstrate that Skeleton Recall Loss shows overall superior performance on existing state-of-the-art topology-aware loss functions. Moreover, it stands as the first loss function designed for computationally manageable thin structure segmentation within the increasingly significant but hitherto unaddressed multi-class context. In essence, Skeleton Recall Loss represents a significant advancement in the field of thin structure segmentation, offering both efficiency and efficacy. The public availability of our code further facilitates its adoption and serves as a foundation for future advancements in this critical area of study.

24. Centerline Boundary Dice Loss for Vascular Segmentation (cbDice)

- Authors: Pengcheng Shi, Jiesi Hu, Yanwu Yang, Zilve Gao, Wei Liu, Ting Ma
- Year: 2024 (MICCAI) · Source: arXiv
- URL: https://arxiv.org/abs/2407.01517
- Relevance: Extends clDice with boundary and radius awareness arXiv — directly addresses the portal-vs-peripheral branch calibre problem in hepatic vessels.

Abstract

Vascular segmentation in medical imaging plays a crucial role in analysing morphological and functional assessments. Traditional methods, like the centerline Dice (clDice) loss, ensure topology preservation but falter in capturing geometric details, especially under translation and deformation. The combination of clDice with traditional Dice loss can lead to diameter imbalance, favoring larger vessels. Addressing these challenges, we introduce the centerline boundary Dice (cbDice) loss function, which harmonizes topological integrity and geometric nuances, ensuring consistent segmentation across various vessel sizes. cbDice enriches the clDice approach by including boundary-aware aspects, thereby improving geometric detail recognition. It matches the performance of the boundary difference over union (B-DoU) loss through a mask-distance-based approach, enhancing traslation sensitivity. Crucially, cbDice incorporates radius information from vascular skeletons, enabling uniform adaptation to vascular diameter changes and maintaining balance in branch growth and fracture impacts. Furthermore, we conducted a theoretical analysis of clDice variants (cl-X-Dice). We validated cbDice's efficacy on three diverse vascular segmentation datasets, encompassing both 2D and 3D, and binary and multi-class segmentation. Particularly, the method integrated with cbDice demonstrated outstanding performance on the MICCAI 2023 TopCoW Challenge dataset. Our code is made publicly available at: https://github.com/PengchengShi1220/cbDice .

Conclusion

In this study, we introduce the cbDice loss, an innovative advancement within the clDice loss framework, further elaborated through a series of cl-X-Dice metrics. These developments specifically cater to the complexities of vascular segmentation in medical imaging. Extensive evaluations across diverse datasets have validated the efficacy of our method in preserving topological integrity, capturing geometric detail, and maintaining balanced diameter representation in vascular segmentation tasks. Comparative analysis against leading models and loss functions have demonstrated the superior capability of the cbDice loss in meeting the intricate demands of vascular segmentation.

25. **A Persistent Homology-Based Topological Loss for CNN-Based Multi-Class Segmentation of CMR**

- Authors: Nick Byrne, James R. Clough, Israel Valverde, Giovanni Montana, Andrew P. King
- Year: 2021 (IEEE TMI) · Source: arXiv
- URL: https://arxiv.org/abs/2107.12689
- Relevance: Efficient parallel cubical-complex implementation of PH losses for multi-class 3D tasks — supports extension of topology-aware training to portal-vs-hepatic-vein multi-class setups.

Abstract

Multi-class segmentation of cardiac magnetic resonance (CMR) images seeks a separation of data into anatomical components with known structure and configuration. The most popular CNN-based methods are optimised using pixel wise loss functions, ignorant of the spatially extended features that characterise anatomy. Therefore, whilst sharing a high spatial overlap with the ground truth, inferred CNN-based segmentations can lack coherence, including spurious connected components, holes and voids. Such results are implausible, violating anticipated anatomical topology. In response, (single-class) persistent homology-based loss functions have been proposed to capture global anatomical features. Our work extends these approaches to the task of multi-class segmentation. Building an enriched topological description of all class labels and class label pairs, our loss functions make predictable and statistically significant improvements in segmentation topology using a CNN-based postprocessing framework. We also present (and make available) a highly efficient implementation based on cubical complexes and parallel execution, enabling practical application within high resolution 3D data for the first time. We demonstrate our approach on 2D short axis and 3D whole heart CMR segmentation, advancing a detailed and faithful analysis of performance on two publicly available datasets.

Conclusion

mage segmentation. In the context of state of the art spatial overlap performance, our novel approach made statistically significant and predictable improvements in label map topology within 2D and 3D tasks. Our approach is theoretically founded, building a multi-class prior as the collection of individual and paired label map topologies. Compared with the naive consideration of singular labels (the natural extension of previous work), the superiority of this scheme is borne out experimentally. Crucially, we adopted a highly efficient algorithmic backbone for the computation of cubical PH, achieving dramatic improvements in execution time and permitting practicable extension to the 3D setting. A careful analysis of both quantitative and qualitative performance allowed us to reflect the limitations of our approach and consider its wider application. Whilst demonstrated in the field of CMR image analysis, our formulation is generalisable to any multi-class segmentation task: we envisage many applications across a diverse range of anatomical and pathological targets.

Liver and liver-tumor segmentation (related domain)

26. The Liver Tumor Segmentation Benchmark (LiTS)

- Authors: Patrick Bilic, Patrick F. Christ, Hongwei Bran Li, Eugene Vorontsov, et al. (incl. Bjoern H. Menze)
- Year: 2019/2023 · Source: arXiv (Medical Image Analysis 2023)
- URL: https://arxiv.org/abs/1901.04056
- Relevance: The definitive liver-CT benchmark; essential context for liver-domain CT segmentation research arXiv and MSD task lineage.

Abstract

In this work, we report the set-up and results of the Liver Tumor Segmentation Benchmark (LiTS), which was organized in conjunction with the IEEE International Symposium on Biomedical Imaging (ISBI) 2017 and the International Conferences on Medical Image Computing and Computer-Assisted Intervention (MICCAI) 2017 and 2018. The image dataset is diverse and contains primary and secondary tumors with varied sizes and appearances with various lesion-to-background levels (hyper-/hypo-dense), created in collaboration with seven hospitals and research institutions. Seventy-five submitted liver and liver tumor segmentation algorithms were trained on a set of 131 computed tomography (CT) volumes and were tested on 70 unseen test images acquired from different patients. We found that not a single algorithm performed best for both liver and liver tumors in the three events. The best liver segmentation algorithm achieved a Dice score of 0.963, whereas, for tumor segmentation, the best algorithms achieved Dices scores of 0.674 (ISBI 2017), 0.702 (MICCAI 2017), and 0.739 (MICCAI 2018). Retrospectively, we performed additional analysis on liver tumor detection and revealed that not all top-performing segmentation algorithms worked well for tumor detection. The best liver tumor detection method achieved a lesion-wise recall of 0.458 (ISBI 2017), 0.515 (MICCAI 2017), and 0.554 (MICCAI 2018), indicating the need for further research. LiTS remains an active benchmark and resource for research, e.g., contributing the liver-related segmentation tasks in http://medicaldecathlon.com/. In addition, both data and online evaluation are accessible via www.lits-challenge.com.

27. **H-DenseUNet: Hybrid Densely Connected UNet for Liver and Tumor Segmentation from CT Volumes**

- Authors: Xiaomeng Li, Hao Chen, Xiaojuan Qi, Qi Dou, Chi-Wing Fu, Pheng-Ann Heng
- Year: 2018 · Source: arXiv + IEEE Xplore (IEEE TMI 37(12))
- URLs: https://arxiv.org/abs/1709.07330 · https://ieeexplore.ieee.org/document/8379359
- Relevance: Classic 2D+3D hybrid for LiTS IEEE XplorearXiv — conceptually close to multiplanar fusion and a strong liver-domain architectural reference.

Abstract

Liver cancer is one of the leading causes of cancer death. To assist doctors in hepatocellular carcinoma diagnosis and treatment planning, an accurate and automatic liver and tumor segmentation method is highly demanded in the clinical practice. Recently, fully convolutional neural networks (FCNs), including 2D and 3D FCNs, serve as the back-bone in many volumetric image segmentation. However, 2D convolutions can not fully leverage the spatial information along the third dimension while 3D convolutions suffer from high computational cost and GPU memory consumption. To address these issues, we propose a novel hybrid densely connected UNet (H-DenseUNet), which consists of a 2D DenseUNet for efficiently extracting intra-slice features and a 3D counterpart for hierarchically aggregating volumetric contexts under the spirit of the auto-context algorithm for liver and tumor segmentation. We formulate the learning process of H-DenseUNet in an end-to-end manner, where the intra-slice representations and inter-slice features can be jointly optimized through a hybrid feature fusion (HFF) layer. We extensively evaluated our method on the dataset of MICCAI 2017 Liver Tumor Segmentation (LiTS) Challenge and 3DIRCADb Dataset. Our method outperformed other state-of-the-arts on the segmentation results of tumors and achieved very competitive performance for liver segmentation even with a single model.

Conclusion

We present an end-to-end training system H-DenseUNet for liver and tumor segmentation from CT volumes, which is a new paradigm to effectively probe high-level representative intra-slice and inter-slice features, followed by optimizing the features through the hybrid feature fusion layer. The architecture gracefully addressed the problems that 2D convolutions ignore the volumetric contexts and 3D convolutions suffer from heavy computational cost. Extensive experiments on the dataset of 2017 LiTS and 3DIRCADb dataset demonstrated the superiority of our proposed H-

DenseUNet. With a singlemodel basis, our method excelled others by a large margin on lesion segmentation and achieved very competitive result on liver segmentation on the LiTS Leaderboard.

28. U-Net: Convolutional Networks for Biomedical Image Segmentation

- Authors: Olaf Ronneberger, Philipp Fischer, Thomas Brox
- Year: 2015 · Source: arXiv
- URL: https://arxiv.org/abs/1505.04597
- Relevance: The seminal encoder-decoder with skip connections — ancestor of every architecture in this project.

Abstract

There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less than a second on a recent GPU. The full implementation (based on Caffe) and the trained networks are available at http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net .

Conclusion

The u-net architecture achieves very good performance on very different biomedical segmentation applications. Thanks to data augmentation with elastic defor1 The authors of this algorithm have submitted 78 different solutions to achieve this result. 2 Data set provided by Dr. Sanjay Kumar. Department of Bioengineering University of California at Berkeley. Berkeley CA (USA) 3 Data set provided by Dr. Gert van Cappellen Erasmus Medical Center. Rotterdam. The Netherland mations, it only needs very few annotated images and has a very reasonable training time of only 10 hours on a NVidia Titan GPU (6 GB). We provide the full Caffe[6]-based implementation and the trained networks4 . We are sure that the u-net architecture can be applied easily to many more tasks.

29. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation

- Authors: Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, Olaf Ronneberger
- Year: 2016 · Source: arXiv
- URL: https://arxiv.org/abs/1606.06650
- Relevance: Volumetric U-Net — the exact 3D_fullres configuration used by nnU-Net v2 in the project.

Abstract

This paper introduces a network for volumetric segmentation that learns from sparsely annotated volumetric images. We outline two attractive use cases of this method: (1) In a semi-automated setup, the user annotates some slices in the volume to be segmented. The network learns from these sparse annotations and provides a dense 3D segmentation. (2) In a fully-automated setup, we assume that a representative, sparsely annotated training set exists. Trained on this data set, the network densely segments new volumetric images. The proposed network extends the previous u-net architecture from Ronneberger et al. by replacing all 2D operations with their 3D counterparts. The implementation performs on-the-fly elastic deformations for efficient data augmentation during training. It is trained end-to-end from scratch, i.e., no pre-trained network is required. We test the performance of the proposed method on a complex, highly variable 3D structure, the Xenopus kidney, and achieve good results for both use cases.

Introduction

Volumetric data is abundant in biomedical data analysis. Annotation of such data with segmentation labels causes difficulties, since only 2D slices can be shown on a computer screen. Thus, annotation of large volumes in a slice-by-slice manner is very tedious. It is inefficient, too, since neighboring slices show almost the same information. Especially for learning based approaches that require a significant amount of annotated data, full annotation of 3D volumes is not an effective way to create large and rich training data sets that would generalize well.

In this paper, we suggest a deep network that learns to generate dense volumetric segmentations, but only requires some annotated 2D slices for training. This network can be used in two different ways as depicted in Fig. 1: the first application case just aims on densification of a sparsely annotated data set; the second learns from multiple sparsely annotated data sets to generalize to new data. Both cases are highly relevant. The network is based on the previous u-net architecture, which consists of a contracting encoder part to analyze the whole image and a successive expanding decoder part to produce a full-resolution segmentation [11]. While the u-net is an entirely 2D architecture, the network proposed in this paper takes 3D volumes as input and processes them with corresponding 3D operations, in particular, 3D convolutions, 3D max pooling, and 3D up-convolutional layers. Moreover, we avoid bottlenecks in the network architecture [13] and use batch normalization [4] for faster convergence. In many biomedical applications, only very few images are required to train a network that generalizes reasonably well. This is because each image already comprises repetitive structures with corresponding variation. In volumetric images, this effect is further pronounced, such that we can train a network on just two volumetric images in order to generalize to a third one. A weighted loss function and special data augmentation enable us to train the network with only few manually annotated slices, i.e., from sparsely annotated training data.

We show the successful application of the proposed method on difficult confocal microscopic data set of the Xenopus kidney. During its development, the Xenopus kidney forms a complex structure [7] which limits the applicability of pre-defined parametric models. First we provide qualitative results to demonstrate the quality of the densification from few annotated slices. These results are supported by quantitative evaluations. We also provide experiments which shows the effect of the number of annotated slices on the performance of our network. The Caffe[5] based network implementation is provided as OpenSource1 .

Conclusion

We have introduced an end-to-end learning method that semi-automatically and fully-automatically segments a 3D volume from a sparse annotation. It offers an accurate segmentation for the highly variable structures of the Xenopus kidney. We achieve an average IoU of 0.863 in 3-fold cross validation experiments for the semi-automated setup. In a fully-automated setup we demonstrate the performance gain of the 3D architecture to an equivalent 2D implementation. The network is trained from scratch, and it is not optimized in any way for this application. We expect that it will be applicable to many other biomedical volumetric segmentation tasks. Its implementation is provided as OpenSource.

30. **V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation**

- Authors: Fausto Milletari, Nassir Navab, Seyed-Ahmad Ahmadi
- Year: 2016 (3DV) · Source: arXiv
- URL: https://arxiv.org/abs/1606.04797
- Relevance: Introduces the **soft Dice loss** arXiv combined with clDice in this project, plus a volumetric residual CNN backbone.

Abstract

Convolutional Neural Networks (CNNs) have been recently employed to solve problems from both the computer vision and medical image analysis fields. Despite their popularity, most approaches are only able to process 2D images while most medical data used in clinical practice consists of 3D volumes. In this work we propose an approach to 3D image segmentation based on a volumetric, fully convolutional, neural network. Our CNN is trained end-to-end on MRI volumes depicting prostate, and learns to predict segmentation for the whole volume at once. We introduce a novel objective function, that we optimise during training, based on Dice coefficient. In this way we can deal with situations where there is a strong imbalance between the number of foreground and background voxels. To cope with the limited number of annotated volumes available for training, we augment the data applying random non-linear transformations and histogram matching. We show in our experimental evaluation that our approach achieves good performances on challenging test data while requiring only a fraction of the processing time needed by other previous methods.

Conclusion

We presented and approach based on a volumetric convolutional neural network that performs segmentation of MRI prostate volumes in a fast and accurate manner. We introduced a novel objective function that we optimise during training based on the Dice overlap coefficient between the predicted segmentation and the ground truth annotation. Our Dice loss layer does not need sample re-weighting when the amount of background and foreground pixels is strongly unbalanced and is indicated for binary segmentation tasks. Although we inspired our architecture to the one proposed in [14], we divided it into stages that learn residuals and, as empirically observed, improve both results and convergence time. Future works will aim at segmenting volumes containing multiple regions in other modalities such as ultrasound and at higher resolutions by splitting the network over multiple GPUs.

31. Attention U-Net: Learning Where to Look for the Pancreas

- Authors: Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, et al.
- Year: 2018 · Source: arXiv
- URL: https://arxiv.org/abs/1804.03999
- Relevance: Attention gates that emphasise small target regions — highly transferable to sparse, thin hepatic vessels.

Abstract

We propose a novel attention gate (AG) model for medical imaging that automatically learns to focus on target structures of varying shapes and sizes. Models trained with AGs implicitly learn to suppress irrelevant regions in an input image while highlighting salient features useful for a specific task. This enables us to eliminate the necessity of using explicit external tissue/organ localisation modules of cascaded convolutional neural networks (CNNs). AGs can be easily integrated into standard CNN architectures such as the U-Net model with minimal computational overhead while increasing the model sensitivity and prediction accuracy. The proposed Attention U-Net architecture is evaluated on two large CT abdominal datasets for multi-class image segmentation. Experimental results show that AGs consistently improve the prediction performance of U-Net across different datasets and training sizes while preserving computational efficiency. The source code for the proposed architecture is publicly available.

Conclusion

In this paper, we presented a novel attention gate model applied to medical image segmentation. Our approach eliminates the necessity of applying an external object localisation model. The proposed approach is generic and modular as such it can be easily applied to image classification and regression problems as in the examples of natural image analysis and machine translation. Experimental results demonstrate that the proposed AGs are highly beneficial for tissue/organ identification and localisation. This is particularly true for variable small size organs such as the pancreas, and similar behaviour is expected for global classification tasks. Training behaviour of the AGs can benefit from transfer learning and multi-stage training schemes. For instance, pre-trained U-Net weights can be used to initialise the attention network, and gates can be trained accordingly in the fine-tuning stage. Similarly, there is a vast body of literature in machine learning exploring different gating architectures. For example, highway networks [7] make use of residual connections around the gate block to allow better gradient backpropagation and slightly softer attention mechanisms. Although our experiments with residual connections have not provided any significant performance improvement, future research will focus on this aspect to obtain a better training behaviour. Lastly, we note that with the advent of improved GPU computation power and memory, larger capacity 3D models can be trained with larger batch sizes without the need for image downsampling. In this way, we would not need to utilise ad-hoc post-processing techniques to further improve the state-of-the-art results. Similarly, the performance of Attention U-Net can be further enhanced by utilising fine resolution input batches without additional heuristics. Lastly, we would like to thank to Salim Arslan and Dan Busbridge for their helpful comments on this work.

32. **TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation**

- Authors: Jieneng Chen, Yongyi Lu, Qihang Yu, Xiangde Luo, Ehsan Adeli, Yan Wang, Le Lu, Alan L. Yuille, Yuyin Zhou
- Year: 2021 · Source: arXiv
- URL: https://arxiv.org/abs/2102.04306
- Relevance: Hybrid CNN-Transformer encoder PubMed CentralarXiv capturing long-range context — helps maintain continuity along elongated vessel branches.

Abstract

Medical image segmentation is an essential prerequisite for developing healthcare systems, especially for disease diagnosis and treatment planning. On various medical image segmentation tasks, the ushaped architecture, also known as U-Net, has become the de-facto standard and achieved tremendous success. However, due to the intrinsic locality of convolution operations, U-Net generally demonstrates limitations in explicitly modeling long-range dependency. Transformers, designed for sequence-to-sequence prediction, have emerged as alternative architectures with innate global self-attention mechanisms, but can result in limited localization abilities due to insufficient low-level details. In this paper, we propose TransUNet, which merits both Transformers and U-Net, as a strong alternative for medical image segmentation. On one hand, the Transformer encodes tokenized image patches from a convolution neural network (CNN) feature map as the input sequence for extracting global contexts. On the other hand, the decoder upsamples the encoded features which are then combined with the high-resolution CNN feature maps to enable precise localization. We argue that Transformers can serve as strong encoders for medical image segmentation tasks, with the combination of U-Net to enhance finer details by recovering localized spatial information. TransUNet achieves superior performances to various competing methods on different medical applications including multi-organ segmentation and cardiac segmentation. Code and models are available at https://github.com/Beckschen/TransUNet .

Conclusion

Transformers are known as architectures with strong innate self-attention mechanisms. In this paper, we present the first study to investigate the usage of Transformers for general medical image segmentation. To fully leverage the power of Transformers, TransUNet was proposed, which not only encodes strong global context by treating the image features as sequences but also well utilizes the low-level CNN features via a u-shaped hybrid architectural design. As an alternative framework to the dominant FCN-based approaches for medical image segmentation, TransUNet achieves superior performances than various competing methods, including CNN-based self-attention methods. Acknowledgements. This work was supported by the Lustgarten Foundation for Pancreatic Cancer Research.

33. UNETR: Transformers for 3D Medical Image Segmentation

- Authors: Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger Roth, Daguang Xu
- Year: 2021 (WACV 2022) · Source: arXiv
- URL: https://arxiv.org/abs/2103.10504
- Relevance: Pure-transformer 3D encoder arXiv evaluated on BTCV and MSD arXiv — commonly benchmarked against nnU-Net.

Abstract

Fully Convolutional Neural Networks (FCNNs) with contracting and expanding paths have shown prominence for the majority of medical image segmentation applications since the past decade. In FCNNs, the encoder plays an integral role by learning both global and local features and contextual representations which can be utilized for semantic output prediction by the decoder. Despite their success, the locality of convolutional layers in FCNNs, limits the capability of learning long-range spatial dependencies. Inspired by the recent success of transformers for Natural Language Processing (NLP) in long-range sequence learning, we reformulate the task of volumetric (3D) medical image segmentation as a sequence-to-sequence prediction problem. We introduce a novel architecture, dubbed as UNEt TRansformers (UNETR), that utilizes a transformer as the encoder to learn sequence representations of the input volume and effectively capture the global multi-scale information, while also following the successful "U-shaped" network design for the encoder and decoder. The transformer encoder is directly connected to a decoder via skip connections at different resolutions to compute the final semantic segmentation output. We have validated the performance of our method on the Multi Atlas Labeling Beyond The Cranial Vault (BTCV) dataset for multiorgan segmentation and the Medical Segmentation Decathlon (MSD) dataset for brain tumor and spleen segmentation tasks. Our benchmarks demonstrate new state-of-the-art performance on the BTCV leaderboard. Code: https://monai.io/research/unetr

Conclusion

This paper introduces a novel transformer-based architecture, dubbed as UNETR, for semantic segmentation of volumetric medical images by reformulating this task as a 1D sequence-to-sequence prediction problem. We proposed to use a transformer encoder to increase the model's capability for learning long-range dependencies and effectively capturing global contextual representation at multiple scales. We validated the effectiveness of UNETR on different volumetric segmentation tasks in CT and MRI modalities. UNETR achieves new state-of-the-art performance in both Standard and

Free Competitions on the BTCV leaderboard for the multi-organ segmentation and outperforms competing approaches for brain tumor and spleen segmentation on the MSD dataset. In conclusion, UNETR has shown the potential to effectively learn the critical anatomical relationships represented in medical images. The proposed method could be the foundation for a new class of transformer-based segmentation models in medical images analysis.

34. **Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images**

- Authors: Ali Hatamizadeh, Vishwesh Nath, Yucheng Tang, Dong Yang, Holger Roth, Daguang Xu
- Year: 2022 · Source: arXiv
- URL: https://arxiv.org/abs/2201.01266
- Relevance: Hierarchical Swin-Transformer encoder TheobjectsarXiv used as the reference transformer baseline in recent hepatic-vessel papers.

Abstract

Semantic segmentation of brain tumors is a fundamental medical image analysis task involving multiple MRI imaging modalities that can assist clinicians in diagnosing the patient and successively studying the progression of the malignant entity. In recent years, Fully Convolutional Neural Networks (FCNNs) approaches have become the de facto standard for 3D medical image segmentation. The popular "U-shaped" network architecture has achieved state-of-the-art performance benchmarks on different 2D and 3D semantic segmentation tasks and across various imaging modalities. However, due to the limited kernel size of convolution layers in FCNNs, their performance of modeling long-range information is sub-optimal, and this can lead to deficiencies in the segmentation of tumors with variable sizes. On the other hand, transformer models have demonstrated excellent capabilities in capturing such longrange information in multiple domains, including natural language processing and computer vision. Inspired by the success of vision transformers and their variants, we propose a novel segmentation model termed Swin UNEt TRansformers (Swin UNETR). Specifically, the task of 3D brain tumor semantic segmentation is reformulated as a sequence to sequence prediction problem wherein multi-modal input data is projected into a 1D sequence of embedding and used as an input to a hierarchical Swin transformer as the encoder. The swin transformer encoder extracts features at five different resolutions by utilizing shifted windows for computing self-attention and is connected to an FCNN-based decoder at each resolution via skip connections. We have participated in BraTS 2021 segmentation challenge, and our proposed model ranks among the top-performing approaches in the validation phase. Code: https://monai.io/research/swin-unetr

Conclusion

In this paper, we introduced Swin UNETR which is a novel architecture for semantic segmentation of brain tumors using multi-modal MRI images. Our proposed model has a U-shaped network design and uses a Swin transformer as the encoder and CNN-based decoder that is connected to the encoder via skip connections at different resolutions. We have validated the effectiveness of our approach by in the BraTS 2021 challenge. Our model ranks among topperforming approaches in the validation phase and demonstrates competitive performance in the testing phase. We believe that Swin UNETR could be the foundation of a new class of transformer-based models with hierarchical encoders for the task of brain tumor segmentation.

35. Self-Supervised Pre-Training of Swin Transformers for 3D Medical Image Analysis

- Authors: Yucheng Tang, Dong Yang, Wenqi Li, Holger R. Roth, Bennett Landman, Daguang Xu, Vishwesh Nath, Ali Hatamizadeh
- Year: 2022 (CVPR) · Source: arXiv
- URL: https://arxiv.org/abs/2111.14791
- Relevance: Demonstrates large-scale self-supervised pretraining of Swin-UNETR boosting MSD/BTCV performance — key comparator for transformer-based hepatic vessel pipelines.

Abstract

Vision Transformers (ViT)s have shown great performance in self-supervised learning of global and local representations that can be transferred to downstream applications. Inspired by these results, we introduce a novel self-supervised learning framework with tailored proxy tasks for medical image analysis. Specifically, we propose: (i) a new 3D transformer-based model, dubbed Swin UNEt TRansformers (Swin UNETR), with a hierarchical encoder for self-supervised pre-training; (ii) tailored proxy tasks for learning the underlying pattern of human anatomy. We demonstrate successful pre-training of the proposed model on 5,050 publicly available computed tomography (CT) images from various body organs. The effectiveness of our approach is validated by fine-tuning the pre-trained models on the Beyond the Cranial Vault (BTCV) Segmentation Challenge with 13 abdominal organs and segmentation tasks from the Medical Segmentation Decathlon (MSD) dataset. Our model is currently the state-of-the-art on the public test leaderboards of both MSD1 and BTCV 2 datasets. Code: https://monai.io/research/swin-unetr.

Conclusion

In this work, we present a novel framework for selfsupervised pre-training of 3D medical images. Inspired by merging feature maps at scales, we built the Swin UNETR by exploiting transformer-encoded spatial representations into convolution-based decoders. By proposing the first transformer-based 3D medical image pre-training, we leverage the power of Swin Transformer encoder for fine-tuning segmentation tasks. Swin UNETR with self-supervised pre-training achieves the state-of-the-art performance on the BTCV multi-organ segmentation challenge and MSD challenge. Particularly, we presented the large-scale CT pre-training with 5,050 volumes, by combining multiple publicly available datasets and diversities of anatomical ROIs.

36. DeepVesselNet: Vessel Segmentation, Centerline Prediction, and Bifurcation Detection in 3-D Angiographic Volumes

- Authors: Giles Tetteh, Velizar Efremov, Nils D. Forkert, Matthias Schneider, Jan Kirschke, Bruno Weber, Claus Zimmer, Marie Piraud, Bjoern H. Menze
- Year: 2018/2020 · Source: arXiv
- URL: https://arxiv.org/abs/1803.09340
- Relevance: Multi-task 3D CNN performing vessel segmentation + centerline + bifurcation detection — a pipeline template directly applicable to hepatic vessels.

Abstract

We present DeepVesselNet, an architecture tailored to the challenges faced when extracting vessel networks or trees and corresponding features in 3-D angiographic volumes using deep learning. We discuss the problems of low execution speed and high memory requirements associated with full 3-D convolutional networks, high-class imbalance arising from the low percentage (less than 3%) of vessel voxels, and unavailability of accurately annotated training data - and offer solutions as the building blocks of DeepVesselNet. First, we formulate 2-D orthogonal cross-hair filters which make use of 3-D context information at a reduced computational burden. Second, we introduce a class balancing cross-entropy loss function with false positive rate correction to handle the highclass imbalance and high false positive rate problems associated with existing loss functions. Finally, we generate synthetic dataset using a computational angiogenesis model capable of generating vascular trees under physiological constraints on local network structure and topology and use these data for transfer learning. DeepVesselNet is optimized for segmenting and analyzing vessels, and we test the performance on a range of angiographic volumes including clinical time-of-flight MRA data of the human brain, as well as synchrotron radiation X-ray tomographic microscopy scans of the rat brain. Our experiments show that, by replacing 3-D filters with cross-hair filters in our network, we achieve over 23% improvement in speed, lower memory footprint, lower network complexity which prevents overfitting and comparable accuracy (with a Cox-Wilcoxon paired sample significance test p-value of 0.07 when compared to full 3-D filters). Our class balancing metric is crucial for training the network and transfer learning with synthetic data is an efficient, robust, and very generalizable approach leading to a network that excels in a variety of angiography segmentation tasks. We show that sub-sampling and max pooling layers may lead to drop in performance in tasks that involve voxel-sized structures and DeepVesselNet architecture which does not use any form of subsampling layer works well for vessel segmentation, centerline prediction and bifurcation detection. We make available our datasets publicly for downloading,

fostering future research, in particular, on centerline prediction and bifurcation detection and serving as one of the first public datasets for brain vessel tree segmentation and analysis.

Conclusion

We present DeepVesselNet, an architecture tailored to the challenges of extracting vessel networks and network features using deep learning. Our experiments in Sections IV-C and IV-D show that the cross-hair filters, which is one of the components of DeepVesselNet, performs comparably well as 3-D filters and, at the same time, improves significantly both speed and memory usage, easing an upscaling to larger data sets. Another component of DeepVesselNet, the introduction of new weights and the FP rate correction discussed in Section III-B, helps in maintaining a good balance between precision and recall during training. This turns out to be crucial for preventing over and under-segmentation problems, which are common problems in vessel segmentation. We also show from our results in Section IV-D that using sub-sampling layers in a network architecture in tasks which includes voxelsized objects can lead to a fall in performance. Finally, we successfully demonstrated in Sections IV-C and IV-D that transfer learning of DeepVesselNet through pretraining on synthetically generated data improves segmentation and detection results, especially in situations where obtaining manually annotated data is a challenge. As future work, we will generalize DeepVesselNet to multiclass vessel tree task, handling vessel segmentation, centerline prediction, and bifurcation detection simultaneously, rather than in three subsequent binary tasks. We also expect that network architectures tailored to our three hierarchically nested classes will improve the performance of the DeepVesselNet. For example by using a multi-level activation approach proposed in [52] or through in a single, but hierarchical approach starting from a base network for vessel segmentation, additional layers for centerline prediction, and a final set of layers for bifurcation detection. The current implementation of cross-hair filters, network architectures and cost function are available on Github at https://github.com/giesekow/deepvesselnet. Datasets can also be downloaded from the wiki page of the same link above. Future extensions to DeepVesselNet, as well as any additional datasets will be made publicly available on Github.

37. Retinal Vessel Segmentation Based on Fully Convolutional Networks

- Authors: Zhengyuan Liu
- Year: 2019 · Source: arXiv
- URL: https://arxiv.org/abs/1911.09915
- Relevance: Benchmarks U-Net/LadderNet on DRIVE/STARE/CHASE_DB1 — patch-based training and thin-vessel connectivity analysis transfer to 3D hepatic vessels.

Abstract

The morphological attributes of retinal vessels, such as length, width, tortuosity and branching pattern and angles, play an important role in diagnosis, screening, treatment, and evaluation of various cardiovascular and ophthalmologic diseases such as diabetes, hypertension and arteriosclerosis. The crucial step before extracting these morphological characteristics of retinal vessels from retinal fundus images is vessel segmentation. In this work, we propose a method for retinal vessel segmentation based on fully convolutional networks. Thousands of patches are extracted from each retinal image and then fed into the network, and data argumentation is applied by rotating extracted patches. Two architectures of fully convolutional networks, U-Net and LadderNet, are used for vessel segmentation. The performance of our method is evaluated on three public datasets: DRIVE, STARE, and CHASE DB1. Experimental results of our method show superior performance compared to recent state-of-the-art methods.

Conclusion

In this work, we presented a method for retinal vessel segmentation using patchbased fully convolutional networks. Patches are extracted from the retinal image and then fed into the networks, and data argumentation is applied by rotating extracted patches. We applied two network architectures for vessel segmentation: U-Net and LadderNet. Experimental results of our method show superior performance compared to recent state-of-the-art methods. The limitation of our vessel segmentation method is the robustness and connectivity of the output vessel segmentation, as some thin vessels are in small, broken bits and not connected to the main vessel tree. The borders of the patch also have some side-effects when generating the segmentation mask for the full image. For future work, more advanced network architectures, such as DenseNet [11] or DeepLabv3+ [2], can be used to further improve the performance and make the segmentation output more robust. Moreover, large-size patches, even full images, can be used as the input to the networks, to give the networks more information from the input and alleviate the size-effects of patches.

38. Coronary Artery Segmentation in Cardiac CT Angiography Using 3D Multi-Channel U-Net

- Authors: Zhuowen Chen, Yibo Xie, Chun Xie, et al.
- Year: 2019 · Source: arXiv
- URL: https://arxiv.org/abs/1907.12246
- Relevance: Uses vesselness filter responses as auxiliary input channels for a 3D U-Net — transferable preprocessing strategy for CT hepatic vessels.

Abstract

Vessel stenosis is a major risk factor in cardiovascular diseases (CVD). To analyze the degree of vessel stenosis for supporting the treatment management, extraction of coronary artery area from Computed Tomographic Angiography (CTA) is regarded as a key procedure. However, manual segmentation by cardiologists may be a timeconsuming task, and present a significant inter-observer variation. Although various computer-aided approaches have been developed to support segmentation of coronary arteries in CTA, the results remain unreliable due to complex attenuation appearance of plaques, which are the cause of the stenosis. To overcome the difficulties caused by attenuation ambiguity, in this paper, a 3D multi-channel U-Net architecture is proposed for fully automatic 3D coronary artery reconstruction from CTA. Other than using the original CTA image, the main idea of the proposed approach is to incorporate the vesselness map into the input of the U-Net, which serves as the reinforcing information to highlight the tubular structure of coronary arteries. The experimental results show that the proposed approach could achieve a Dice Similarity Coefficient (DSC) of 0.8 in comparison to around 0.6 attained by previous CNN approaches.

Conclusion

Other approaches in deep learning for coronary artery segmentation show relatively limited performance[10] [11]. In comparison, the proposed approach achieves 0.8 DSC performance. The reason might be due to U-net having the same output size as the input image, therefore it is much more suitable than normal CNN in segmentation task. Secondly, the multi-channel concept, allowed the model to have tubular structural information during training, aiding the network to learn more effectively. In Kirişli et al. [14], ground-truth is the average region of the three observers, so there would be three performances for each observer. It is worth noting that the best observer has only 0.79 performance in DSC, thus the algorithm has shown to outperform the observer's performance.

39. Coronary Artery Centerline Extraction in Cardiac CT Angiography Using a CNN-Based Orientation Classifier

- Authors: Jelmer M. Wolterink, Robbert W. van Hamersvelt, Max A. Viergever, Tim Leiner, Ivana Išgum
- Year: 2018 · Source: arXiv
- URL: https://arxiv.org/abs/1810.03143
- Relevance: 3D dilated CNN tracker for tubular centerlines — relevant for post-hoc centerline-based topology validation.

Abstract

Coronary artery centerline extraction in cardiac CT angiography (CCTA) images is a prerequisite for evaluation of stenoses and atherosclerotic plaque. In this work, we propose an algorithm that extracts coronary artery centerlines in CCTA using a convolutional neural network (CNN). In the proposed method, a 3D dilated CNN is trained to predict the most likely direction and radius of an artery at any given point in a CCTA image based on a local image patch. Starting from a single seed point placed manually or automatically anywhere in a coronary artery, a tracker follows the vessel centerline in two directions using the predictions of the CNN. Tracking is terminated when no direction can be identified with high certainty. The CNN is trained using manually annotated centerlines in training images. No image preprocessing is required, so that the process is guided solely by the local image values around the tracker's location. The CNN was trained using a training set consisting of 8 CCTA images with a total of 32 manually annotated centerlines provided in the MICCAI 2008 Coronary Artery Tracking Challenge (CAT08). Evaluation was performed within the CAT08 challenge using a test set consisting of 24 CCTA test images in which 96 centerlines were extracted. The extracted centerlines had an average overlap of 93.7% with manually annotated reference centerlines. Extracted centerline points were highly accurate, with an average distance of 0.21 mm to reference centerline points. Based on these results the method ranks third among 25 publicly evaluated methods in CAT08. In a second test set consisting of 50 CCTA scans acquired at our institution (UMCU), an expert placed 5,448 markers in the coronary arteries, along with radius measurements. Each marker was used as a seed point to extract a single centerline, which was compared to the other markers placed by the expert. This showed strong correspondence between extracted centerlines and manually placed markers. In a third test set containing 36 CCTA scans from the MICCAI 2014 Challenge on Automatic Coronary Calcium Scoring (orCaScore), fully automatic seeding and centerline extraction was evaluated using a segment-wise analysis. This showed that the algorithm is able to fully-automatically extract on average 92% of clinically relevant coronary artery segments. Finally, the limits of agreement between reference and automatic artery radius measurements were

found to be below the size of one voxel in both the CAT08 dataset and the UMCU dataset. Extraction of a centerline based on a single seed point required on average 0.4 ± 0.1 s and fully automatic coronary tree extraction required around 20 s. The proposed method is able to accurately and efficiently determine the direction and radius of coronary arteries based on information derived directly from the image data. The method can be trained with limited training data, and once trained allows fast automatic or interactive extraction of coronary artery trees from CCTA images.

Conclusion

A deep learning-based method for coronary artery centerline extraction has been proposed. The results show that a convolutional neural network can learn to simultaneously determine the direction of coronary artery centerlines and the radius of the coronary lumen with high speed and accuracy.

40. Cerebrovascular Network Segmentation on MRA Images with Deep Learning

- Authors: Pedro Sanches, Cécile Meyer, Vincent Vigneron, Hichem Maaref
- Year: 2018 · Source: arXiv
- URL: https://arxiv.org/abs/1812.01752
- Relevance: Studies loss functions for sparse 3D vessel targets — same class-imbalance regime as hepatic vessels in CT.

Abstract

Deep learning has been shown to produce state of the art results in many tasks in biomedical imaging, especially in segmentation. Moreover, segmentation of the cerebrovascular structure from magnetic resonance angiography is a challenging problem because its complex geometry and topology have a large inter-patient variability. Therefore, in this work, we present a convolutional neural network approach for this problem. Particularly, a new network topology inspired by the U-net 3D and by the Inception modules, entitled Uception. In addition, a discussion about the best objective function for sparse data also guided most choices during the project. State of the art models are also implemented for a comparison purpose and final results show that the proposed architecture has the best performance in this particular context.

Conclusion

The goal of this work was to segment the cerebrovascular anatomical structure in magnetic resonance angiographic images using deep learning methods. Inspired by the inception architecture, we proposed an U-net like 3D architecture with blocks arranged in a more sparse manner, entitled Uception. Experiments on our dataset showed that our model has a better performance than the original U-net. The choice of the loss function is crucial in this case due to the sparseness of the data. Normal cross-entropy functions won't converge for this problem, so optimizing the Dice coefficient appeared as a good compromise. This is due to the fact that the Dice coefficient does not take into account the total number of voxels in the data. Therefore the huge difference between the amount of voxels from the background and from the anatomical structure does not contribute to the objective function. Evaluating these results is challenging due to the lack of precision of the available ground-truth. Moreover, the tubular structures obtained in the segmentation can be similar to those in ground-truth but with a small shift. It doesn't affect much the anatomical fidelity but really decreases Dice coefficients values. Therefore, the average Hausdorff's distance used in this case shows that our results are really close to the ground truths. In further works we plan to improve the results by adding in the neural network some anatomical knowledge of the cerebral vasculature. This knowledge could be based on a cerebrovascular deformable atlas, as in [14].

41. Automatic Cerebral Vessel Extraction in TOF-MRA Using Deep Learning

- Authors: V. de Vos, K. M. Timmins, I. C. van der Schaaf, Y. Ruigrok, B. K. Velthuis, H. J. Kuijf
- Year: 2021 · Source: arXiv
- URL: https://arxiv.org/abs/2101.09253
- Relevance: Compares 2D and 3D U-Nets plus augmentation sweeps on thin 3D vessels — informs the multiplanar design trade-offs.

Abstract

Deep learning approaches may help radiologists in the early diagnosis and timely treatment of cerebrovascular diseases. Accurate cerebral vessel segmentation of Time-of-Flight Magnetic Resonance Angiographs (TOFMRAs) is an essential step in this process. This study investigates deep learning approaches for automatic, fast and accurate cerebrovascular segmentation for TOF-MRAs. The performance of several data augmentation and selection methods for training a 2D and 3D U-Net for vessel segmentation was investigated in five experiments: a) without augmentation, b) Gaussian blur, c) rotation and flipping, d) Gaussian blur, rotation and flipping and e) different input patch sizes. All experiments were performed by patch-training both a 2D and 3D U-Net and predicted on a test set of MRAs. Ground truth was manually defined using an interactive threshold and region growing method. The performance was evaluated using the Dice Similarity Coefficient (DSC), Modified Hausdorff Distance and Volumetric Similarity, between the predicted images and the interactively defined ground truth. The segmentation performance of all trained networks on the test set was found to be good, with DSC scores ranging from 0.72 to 0.83. Both the 2D and 3D U-Net had the best segmentation performance with Gaussian blur, rotation and flipping compared to other experiments without augmentation or only one of those augmentation techniques. Additionally, training on larger patches or slices gave optimal segmentation results. In conclusion, vessel segmentation can be optimally performed on TOF-MRAs using a trained 3D U-Net on larger patches, where data augmentation including Gaussian blur, rotation and flipping was performed on the training data.

Conclusion

In conclusion, our study found that a 3D U-Net trained on patches of size 64x64x64 voxels augmented using Gaussian blur, rotation and flipping performs optimally for vessel segmentation from TOF-MRAs.

42. Ensembles of Multiple Models and Architectures for Robust Brain Tumour Segmentation (EMMA)

- Authors: Konstantinos Kamnitsas, Wenjia Bai, Enzo Ferrante, Steven McDonagh, Matthew Sinclair, Nick Pawlowski, Martin Rajchl, et al.
- Year: 2017 (BrainLes@MICCAI) · Source: arXiv
- URL: https://arxiv.org/abs/1711.01468
- Relevance: BraTS-2017 winner and canonical reference on ensembling heterogeneous 3D CNNs — justifies the majority-voting design.

Abstract

Deep learning approaches such as convolutional neural nets have consistently outperformed previous methods on challenging tasks such as dense, semantic segmentation. However, the various proposed networks perform differently, with behaviour largely influenced by architectural choices and training settings. This paper explores Ensembles of Multiple Models and Architectures (EMMA) for robust performance through aggregation of predictions from a wide range of methods. The approach reduces the influence of the meta-parameters of individual models and the risk of overfitting the configuration to a particular database. EMMA can be seen as an unbiased, generic deep learning model which is shown to yield excellent performance, winning the first position in the BRATS 2017 competition among 50+ participating teams.

Conclusion

Neural networks have been proven very potent, yet imperfect estimators, often making unpredictable errors. Biomedical applications are reliability-critical however. For this reason we first concentrate on improving robustness. Towards this goal we introduced EMMA, an ensemble of widely varying CNNs. By combining a heterogeneous collection of networks we construct a model that is insensitive to independent failures of CNN components and thus generalises well (Fig. 7). We also introduced the new perspective of ensembling for objectiveness. By marginalizing out via ensembling the biased behaviour introduced by configuration choices, EMMA is a model more fit for objective analysis. Even though the individual networks have straight-forward architectures and were not optimized for the task, EMMA won the first position in the final testing stage of BRATS 2017 competition among 50+ teams, indicating strong generalisation. By being robust to suboptimal configurations of its components, EMMA may offer re-usability on different tasks, which we aim to explore in the future. EMMA could also be useful in unbiased investigation of factors such as sensitivity of CNNs to different sources of domain shift that is strongly affecting large-scale studies [30], or estimating amount of training data required for a task. Finally, EMMA's uncertainty could serve as a more objective measure of what type of patients or tumours are most challenging to learn.

43. Weighted Ensemble of Deep Learning Models Based on Comprehensive Learning Particle Swarm Optimization for Medical Image Segmentation

- Authors: Truong Dang, Tien Thanh Nguyen, Alan Wee-Chung Liew, Eyad Elyan
- Year: 2021 · Source: IEEE Xplore (IJCNN)
- URL: https://ieeexplore.ieee.org/document/9504929
- Relevance: IEEE-indexed weighted-fusion ensemble that beats single models — a natural comparator to plain voxel-level majority voting.

Abstract

In recent years, deep learning has rapidly become a method of choice for segmentation of medical images. Deep neural architectures such as UNet and FPN have achieved high performances on many medical datasets. However, medical image analysis algorithms are required to be reliable, robust, and accurate for clinical applications which can be difficult to achieve for some single deep learning methods. In this study, we introduce an ensemble of classifiers for semantic segmentation of medical images. The ensemble of classifiers here is a set of various deep learning-based classifiers, aiming to achieve better performance than using a single classifier. We propose a weighted ensemble method in which the weighted sum of segmentation outputs by classifiers is used to choose the final segmentation decision. We use a swarm intelligence algorithm namely Comprehensive Learning Particle Swarm Optimization to optimize the combining weights. Dice coefficient, a popular performance metric for image segmentation, is used as the fitness criteria. Experiments conducted on some medical datasets of the CAMUS competition on cardiographic image segmentation show that our method achieves better results than both the constituent segmentation models and the reported model of the CAMUS competition.

44. Two-Layer Ensemble of Deep Learning Models for Medical Image Segmentation

- Authors: Truong Dang, Tien Thanh Nguyen, John McCall, Eyad Elyan, Carlos F. Moreno-García
- Year: 2021 · Source: arXiv
- URL: https://arxiv.org/abs/2104.04809
- Relevance: Weighted ensemble of heterogeneous CNNs with learned weights — useful design alternative to majority voting across planes.

Abstract

In recent years, deep learning has rapidly become a method of choice for the segmentation of medical images. Deep Neural Network (DNN) architectures such as UNet have achieved state-of-the-art results on many medical datasets. To further improve the performance in the segmentation task, we develop an ensemble system which combines various deep learning architectures. We propose a two-layer ensemble of deep learning models for the segmentation of medical images. The prediction for each training image pixel made by each model in the first layer is used as the augmented data of the training image for the second layer of the ensemble. The prediction of the second layer is then combined by using a weights-based scheme in which each model contributes differently to the combined result. The weights are found by solving linear regression problems. Experiments conducted on two popular medical datasets namely CAMUS and Kvasir-SEG show that the proposed method achieves better results concerning two performance metrics (Dice Coefficient and Hausdorff distance) compared to some well-known benchmark algorithms.

Conclusion

In this paper, we presented a two-layer ensemble of deep learning models for segmentation of medical images. The key idea is to use the probability prediction by the constituent models in the first layer as augmented data for the second layer. The output probability prediction by the the second layer is combined by using a weight-based scheme which is not only a effective combiner but also computational efficient. The weights are found by solving a linear regression problem associated with each class label. Our results on two benchmark datasets show that the proposed ensemble method is able to combine the strengths and mitigate the drawbacks of the constituent segmentation methods, resulting in an overall improvement

45. MLV2-Net: Rater-Based Majority-Label Voting for Consistent Meningeal Lymphatic Vessel Segmentation

- Authors: Robin Weiler, Florian Kofler, Bastian Wittmann, Ivan Ezhov, et al.
- Year: 2024 · Source: arXiv
- URL: https://arxiv.org/abs/2411.08537
- Relevance: Recent nnU-Net-based vessel pipeline that uses **weighted majority-label voting** — direct methodological analogue to this project.

Abstract

Meningeal lymphatic vessels (MLVs) are responsible for the drainage of waste products from the human brain. An impairment in their functionality has been associated with aging as well as brain disorders like multiple sclerosis and Alzheimer's disease. However, MLVs have only recently been described for the first time in magnetic resonance imaging (MRI), and their ramified structure renders manual segmentation particularly difficult. Further, as there is no consistent notion of their appearance, human-annotated MLV structures contain a high inter-rater variability that most automatic segmentation methods cannot take into account. In this work, we propose a new rateraware training scheme for the popular nnUNet model, and we explore rater-based ensembling strategies for accurate and consistent segmentation of MLVs. This enables us to boost nnU-Net's performance while obtaining explicit predictions in different annotation styles and a rater-based uncertainty estimation. Our final model, MLV2 -Net, achieves a Dice similarity coefficient of 0.806 with respect to the human reference standard. The model further matches the human inter-rater reliability and replicates age-related associations with MLV volume.

Conclusion

In summary, we presented the first automatic method for MLV segmentation from 3D FLAIR imaging. Our model, MLV2 -Net, outperformed state-of-the-art baselines by embracing the styles of all annotators involved in the creation of the training set. In contrast to most segmentation methods, MLV2 -Net provides a rater-based uncertainty estimation. Together with the derived theoretical bounds on the segmented volume, we expect MLV2 -Net to be a valuable tool for clinical researchers that study the glymphatic system. Yet, the technical contributions and code are generic and could be beneficial for other applications as well.

46. The Medical Segmentation Decathlon

- Authors: Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A. Landman, Geert Litjens, Bjoern Menze, Olaf Ronneberger, Ronald M. Summers, Bram van Ginneken, et al.
- Year: 2022 (Nature Communications) / 2021 arXiv · Source: arXiv
- URL: https://arxiv.org/abs/2106.05735
- Relevance: **Primary dataset citation** — formally introduces MSD including Task08 Hepatic Vessel (443 portal-venous CTs with vessel + tumour labels).

Abstract

International challenges have become the de facto standard for comparative assessment of image analysis algorithms given a specific task. Segmentation is so far the most widely investigated medical image processing task, but the various segmentation challenges have typically been organized in isolation, such that algorithm development was driven by the need to tackle a single specific clinical problem. We hypothesized that a method capable of performing well on multiple tasks will generalize well to a previously unseen task and potentially outperform a custom-designed solution. To investigate the hypothesis, we organized the Medical Segmentation Decathlon (MSD)—a biomedical image analysis challenge, in which algorithms compete in a multitude of both tasks and modalities. The underlying data set was designed to explore the axis of difficulties typically encountered when dealing with medical images, such as small data sets, unbalanced labels, multi-site data and small objects. The MSD challenge confirmed that algorithms with a consistent good performance on a set of tasks preserved their good average performance on a different set of previously unseen tasks. Moreover, by monitoring the MSD winner for two years, we found that this algorithm continued generalizing well to a wide range of other clinical problems, further confirming our hypothesis. Three main conclusions can be drawn from this study: (1) state-of-the-art image segmentation algorithms are mature, accurate, and generalize well when retrained on unseen tasks; (2) consistent algorithmic performance across multiple tasks is a strong surrogate of algorithmic generalizability; (3) the training of accurate AI segmentation models is now commoditized to non AI experts.

Conclusion

Machine-learning based semantic segmentation algorithms are becoming increasingly general purpose and accurate, but have historically required significant field-specific expertise to use. The MSD challenge was set up to investigate how accurate fully-automated image segmentation learning methods can be on a plethora of tasks with different types of task complexity. Results from the MSD challenge have demonstrated that fully automated methods can now achieve state-of-the-art performance without the need for manual parameter optimisation, even when applied to previously unseen tasks. A central hypothesis of the MSD challenge— that an algorithm which works well and automatically on several tasks should also work well on other unseen tasks—has been validated among the challenge participants and across tasks. This hypothesis was further corroborated by monitoring the generalizability of the winning method in the two years following the challenge, where we found that nnU-Net achieved state-of-the-art performance on many tasks including against task-optimized networks. While it is important to note that many classic semantic segmentation problems (e.g. domain shift and label accuracy) remain, and that methodological progress (e.g. NAS and better heuristics) will continue pushing the boundaries of algorithmic performance and generalizability, the MSD challenge has demonstrated that the training of accurate semantic segmentation networks can now be fully automated. This commoditization of semantic segmentation methods has the potential to allow non machine learning experts (e.g. clinicians, medical physicists,and applied scientists) to better contribute to, and possibly even independently develop, these techniques.

47. A Large Annotated Medical Image Dataset for the Development and Evaluation of Segmentation Algorithms

- Authors: Amber L. Simpson, Michela Antonelli, Spyridon Bakas, Michel Bilello, Keyvan Farahani, Bram van Ginneken, et al.
- Year: 2019 · Source: arXiv
- URL: https://arxiv.org/abs/1902.09063
- Relevance: The official dataset-release companion paper for MSD — the recommended citation for the Task08 data used in this project.

Abstract

Semantic segmentation of medical images aims to associate a pixel with a label in a medical image without human initialization. The success of semantic segmentation algorithms is contingent on the availability of high-quality imaging data with corresponding labels provided by experts. We sought to create a large collection of annotated medical image datasets of various clinically relevant anatomies available under open source license to facilitate the development of semantic segmentation algorithms. Such a resource would allow: 1) objective assessment of general-purpose segmentation methods through comprehensive benchmarking and 2) open and free access to medical image data for any researcher interested in the problem domain. Through a multi-institutional effort, we generated a large, curated dataset representative of several highly variable segmentation tasks that was used in a crowd-sourced challenge – the Medical Segmentation Decathlon held during the 2018 Medical Image Computing and Computer Aided Interventions Conference in Granada, Spain. Here, we describe these ten labeled image datasets so that these data may be effectively reused by the research community.

48. Restoring Connectivity in Vascular Segmentations Using a Learned Post-Processing Model

- Authors: Sophie Carneiro-Esteves, Antoine Vacavant, Odyssée Merveille
- Year: 2024 (MICCAI) · Source: arXiv
- URL: https://arxiv.org/abs/2404.10506
- Relevance: Learned residual U-Net post-processor that reconnects fragmented vessel components using β_0 error — plug-in module for refining nnU-Net majority-voting outputs.

Abstract

Accurate segmentation of vascular networks is essential for computer-aided tools designed to address cardiovascular diseases. Despite more than thirty years of research, it remains a challenge to obtain vascular segmentation results that preserve the connectivity of the underlying vascular network. Yet connectivity is one of the key features of these tools. In this work, we propose a post-processing algorithm aiming to reconnect vascular structures that have been disconnected by a segmentation algorithm. Connectivity being a complex property to model explicitly, we propose to learn this geometric feature either through synthetic data or annotations of the application of interest. The resulting post-processing model can be used on the output of any supervised or unsupervised vascular segmentation algorithm. We show that this postprocessing effectively restores the connectivity of vascular networks both in 2D and 3D images, leading to improved overall segmentation results.

Conclusion

In this article, we introduced a novel vascular segmentation post-processing to favor vascular network connectivity. This post-processing can be used in an unsupervised or supervised context depending on the availability of vascular annotations on the dataset of interest. We conducted an extensive validation of our approach both in 2D and 3D and showed that our post-processing is robust to the size of disconnections, converges to a reconnected result when used iteratively, and significantly improves the connectivity of segmentation results. We also compared our approach to a recent reconnecting post-processing algorithm, demonstrating that our method reconnects significantly more vessels and does so in a more realistic manner. However, our approach is purely based on the vessel's geometric properties in binary segmentations and thus some false reconnections may appear. Future works include taking into account the intensities of the underlying image to avoid these false reconnections.

Graph Convolution Based Cross-Network Multi-Scale Feature Fusion for Deep Vessel Segmentation

- Authors: Bo Wang, Fan Li, Dongdong Liu, et al.
- Year: 2023 · Source: arXiv
- URL: https://arxiv.org/abs/2301.02393
- Relevance: Combines a CNN and a graph-convolutional U-Net operating on supervoxel graphs — exemplar graph-based refinement architecture for 3D CT vessels.

Abstract

Vessel segmentation is widely used to help with vascular disease diagnosis. Vessels reconstructed using existing methods are often not sufficiently accurate to meet clinical use standards. This is because 3D vessel structures are highly complicated and exhibit unique characteristics, including sparsity and anisotropy. In this paper, we propose a novel hybrid deep neural network for vessel segmentation. Our network consists of two cascaded subnetworks performing initial and refined segmentation respectively. The second subnetwork further has two tightly coupled components, a traditional CNN-based U-Net and a graph U-Net. Cross-network multi-scale feature fusion is performed between these two U-shaped networks to effectively support high-quality vessel segmentation. The entire cascaded network can be trained from end to end. The graph in the second subnetwork is constructed according to a vessel probability map as well as appearance and semantic similarities in the original CT volume. To tackle the challenges caused by the sparsity and anisotropy of vessels, a higher percentage of graph nodes are distributed in areas that potentially contain vessels while a higher percentage of edges follow the orientation of potential nearby vessels. Extensive experiments demonstrate our deep network achieves state-of-the-art 3D vessel segmentation performance on multiple public and in-house datasets.

Conclusion

In this paper, we have presented a cascaded deep neural network for vessel segmentation on CTA images. Our approach represents a new paradigm for modeling the structural information of 3D vessels using deep neural networks through the interaction between a pair of CNN-based U-Net and graph U-Net. By fusing the features across these two types of networks, our method successfully tackles the challenges brought up by the sparsity and anisotropy of vessel structures. Extensive experiments on both public and in-house datasets verify the superiority and effectiveness of our method. By constructing a vessel graph to complement CNNs, our method not only outperforms baseline methods but also achieves the state-of-the-art performance with DICE 89.91/94.8/94.5 on the ASOCA/ACA/HNA datasets, respectively. Our proposed framework provides a stronger spatial structure representation by learning 3D vessel connectivity priors. Our future work includes 1) building a more powerful graph neural network to enhance message passing in our crossnetwork feature fusion module, 2) investigating better graph construction methods by exploiting more domain knowledge from medical experts, and 3) building a high-quality annotated dataset and a friendly open-source code base for 3D vessel segmentation tasks.

50. VSR-Net: Vessel-like Structure Rehabilitation Network with Graph Clustering

- Authors: Haili Ye, Xiaoqing Zhang, Yan Hu, Huazhu Fu, Jiang Liu
- Year: 2024 (IEEE TIP) · Source: arXiv
- URL: https://arxiv.org/abs/2312.13116
- Relevance: Curvilinear clustering + merging post-hoc refinement that repairs subsection ruptures — directly deployable after nnU-Net + clDice inference.

Abstract

The morphologies of vessel-like structures, such as blood vessels and nerve fibres, play significant roles in disease diagnosis, e.g., Parkinson's disease. Deep network-based refinement segmentation methods have recently achieved promising vessellike structure segmentation results. There are still two challenges: (1) existing methods have limitations in rehabilitating subsection ruptures in segmented vessel-like structures; (2) they are often overconfident in predicted segmentation results. To tackle these two challenges, this paper attempts to leverage the potential of spatial interconnection relationships among subsection ruptures from the structure rehabilitation perspective. Based on this, we propose a novel Vessel-like Structure Rehabilitation Network (VSR-Net) to rehabilitate subsection ruptures and improve the model calibration based on coarse vessel-like structure segmentation results. VSR-Net first constructs subsection rupture clusters with Curvilinear Clustering Module (CCM). Then, the welldesigned Curvilinear Merging Module (CMM) is applied to rehabilitate the subsection ruptures to obtain the refined vessellike structures. Extensive experiments on five 2D/3D medical image datasets show that VSR-Net significantly outperforms state-of-the-art (SOTA) refinement segmentation methods with lower calibration error. Additionally, we provide quantitative analysis to explain the morphological difference between the rehabilitation results of VSR-Net and ground truth (GT), which is smaller than SOTA methods and GT, demonstrating that our method better rehabilitates vessel-like structures by restoring subsection ruptures.

Conclusion

In this paper, we provide a novel structure rehabilitation perspective to understand and tackle subsection rupture problems in vessel-like structure segmentation tasks by reconsidering it as the vessel-like structure rehabilitation task. Our proposed VSR-Net can effectively rehabilitate subsection ruptures and improve the confidence calibration under this view based on the coarse vessel-like structure segmentation results, by exploring spatial interconnection relationships among subsection ruptures with graph clustering. The extensive experiments on five 2D/3D vessel-like structure datasets manifest the effectiveness of our method. To further demonstrate the generalization and reliability of our VSR-Net qualitatively and quantitatively, we conduct sound experiments to prove our method is capable of keeping a promising balance among vessel-like structure segmentation results, vessel-like structure rehabilitation results, and confidence calibration of deep networks, keeping consistent with our motivation. In future work, we plan to extend the proposed method to unsupervised vessel-like structure rehabilitation tasks and provide more theoretical analysis.